

Mathematical Simulation of Blood Flow

A Literature Review and Idealized Stenosis Study

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Abstract.

This report reviews the mathematical and numerical setting for simulating idealized stenotic blood flow and frames a controlled reduced-order stenosis study. The literature-review portion fixes the continuum, constitutive, boundary-data, and model-hierarchy vocabulary needed to move from three-dimensional (3D) Navier–Stokes and generalized-Newtonian models to reduced 0D/1D/2D and coupled hemodynamic formulations. Pressure-ratio quantities are treated as computed model outputs unless an external clinical measurement comparison is explicitly introduced. The numerical portion reports a $T = 1$ comparison between the selected 1D solver and resolved 3D velocity data for two idealized stenosis cases, using section-mean and radial-profile velocity diagnostics.

Keywords:

computational hemodynamics, blood flow simulation, stenotic vessels, Navier–Stokes, reduced-order models, idealized stenosis

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1 Introduction and Report Scope

Coronary and arterial stenoses motivate a basic computational question: how do geometry, wall properties, rheology, and boundary data determine the state of blood flow through an idealized narrowed vessel? Clinical pressure-ratio measurements motivate the question because anatomical narrowing alone does not determine functional obstruction. This report, however, is a mathematical and numerical study, not a clinical validation study.

The report fixes the vocabulary needed to move from continuum assumptions, to the three-dimensional incompressible Navier–Stokes and generalized-Newtonian settings, to reduced 0D/1D/2D hemodynamic model classes, and finally to a selected one-dimensional numerical study of an idealized stenosis. Its contribution is twofold: it organizes the relevant mathematical and numerical literature, and it frames a controlled computation for a smooth idealized stenosed vessel.

Pressure-ratio quantities in this report are computed model outputs, not clinical FFR or CT-FFR measurements, unless an external measurement comparison is explicitly introduced. Convention C.2 in Appendix C records the pressure-reference, gauge-pressure, high-flow-window, and admissibility conventions used for any reduced pressure-ratio output.

Literature Review Roadmap

1. Section 1.1 fixes the continuum, kinematic, balance-law, and constitutive vocabulary used later in the report.
2. Section 1.2 identifies the governing equations, boundary-data vocabulary, and model hierarchy.
3. Later literature-review sections use this vocabulary to compare classical finite-difference, finite-volume, and finite-element approaches for hemodynamic models.
4. The numerical-study portion then applies the selected reduced model to a controlled idealized stenosis computation.

1.1 Continuum and Constitutive Assumptions

This section moves from the macroscopic scale assumption, to continuum field variables, to blood-dynamics kinematics, to balance-law vocabulary, and finally to Newtonian versus generalized-Newtonian stress closures.

1.1.1 Physiological Scale Boundary

The present work is posed at the macrocirculatory arterial scale. At this scale, blood is commonly modeled as a viscous fluid whose bulk behavior can be described by continuum mechanics. The goal is not to resolve

the effects of individual blood cells, platelets, or plasma constituents but rather to predict macroscopic observables such as pressure, flow rate, pressure drop, and pressure ratio. [1, Sec. 1.1.2, pp. 9–10] [1, Sec. 1.1.3, pp. 12–14].

1.1.2 Blood as a Continuum

Consider a time-dependent family of spatial fluid domains. For a final time $T_{\text{final}} > 0$, use $I_T := (0, T_{\text{final}})$ for interior-time PDE statements and $\bar{I}_T := [0, T_{\text{final}}]$ for initial data and flow maps. Define

$$\begin{cases} \Omega_B(t) := \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{x} \text{ lies inside the vessel at time } t\} \\ \mathcal{Q}_T := \{(t, \mathbf{x}) : t \in I_T, \mathbf{x} \in \Omega_B(t)\}, \end{cases}$$

where the reference configuration is denoted $\Omega_B(0)$, the current configuration is $\Omega_B(t)$, and the space-time fluid domain is $\mathcal{Q}_T$. This domain notation is the one collected in Appendix B; boundary and function-space conventions are summarized in Appendix D. In this report, the continuum assumption is a large-vessel idealization: individual blood cells, platelets, and plasma constituents are not resolved, and blood velocity, pressure, density, and Cauchy stress are represented by sufficiently regular fields on $\mathcal{Q}_T$. Cellular-scale effects may enter only through later constitutive choices [2, Part I, Ch. 1, Secs. 1.1–1.2, pp. 4–5] [1, Secs. 1.1.3 and 3.2, pp. 12–14 and 46–47]. The continuum setup is as follows: Assumption 1.1, Remark 1.2, Definition 1.3, Definitions 1.5–1.6, Definitions 1.8–1.9, and Definitions 1.13–1.14 fix the assumptions used in the model hierarchy.

1.1.3 Blood Dynamics

This subsection fixes the kinematic vocabulary used by the later continuum and reduced-order models.

Assumption 1.1 (Blood-dynamics kinematic setting). For $t \in \bar{I}_T$, blood motion is represented by an orientation-preserving flow map $\mathcal{L}_t : \Omega_B(0) \rightarrow \Omega_B(t)$ with $\det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t > 0$. The family $\mathcal{L}_t$ has the time and space regularity needed to differentiate $\nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$, define material derivatives, and apply Reynolds transport to material volumes $V(t) = \mathcal{L}_t(V(0))$ with Lipschitz boundary. It carries each material label $\boldsymbol{\xi}$ to its current location $\mathbf{x}(t, \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field $\mathbf{u} : \mathcal{Q}_T \rightarrow \mathbb{R}^3$ satisfies

$$\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \mathbf{u}(t, \mathcal{L}_t(\boldsymbol{\xi})).$$

The accompanying continuum fields are pressure $p : \mathcal{Q}_T \rightarrow \mathbb{R}$, density $\rho : \mathcal{Q}_T \rightarrow \mathbb{R}^+$, and Cauchy stress $\mathbf{T} : \mathcal{Q}_T \rightarrow \mathbb{R}^{3 \times 3}$.

Figure 1 illustrates this Lagrangian–Eulerian notation.

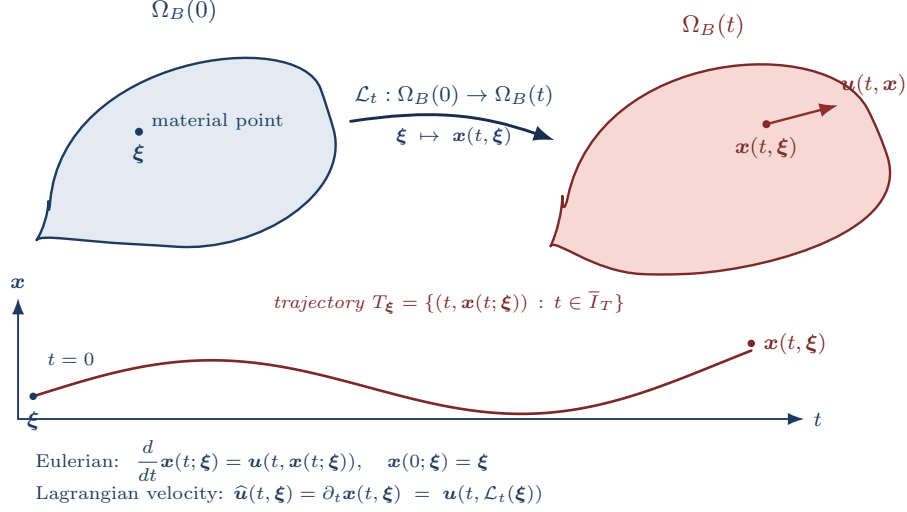


Figure 1: Lagrangian and Eulerian descriptions. The flow map $\mathcal{L}_t : \Omega_B(0) \rightarrow \Omega_B(t)$ carries the material label $\xi \in \Omega_B(0)$ to its current position $\mathbf{x}(t, \xi) \in \Omega_B(t)$. The Eulerian velocity $\mathbf{u}(t, \mathbf{x})$ is the velocity of whichever particle currently sits at $(t, \mathbf{x})$; the Lagrangian velocity is $\hat{\mathbf{u}}(t, \xi) = \partial_t \mathbf{x}(t, \xi) = \mathbf{u}(t, \mathcal{L}_t(\xi))$, and trajectories T_ξ are generated by the velocity field $\mathbf{u}$.

Remark 1.2 (Eulerian vs. Lagrangian). The Eulerian description views fields such as $\phi(t, \mathbf{x})$ or $\mathbf{u}(t, \mathbf{x})$ at fixed spatial locations. The Lagrangian description follows material particles ξ along trajectories generated by the velocity field and views fields as functions of the material label ξ and time t , such as $\hat{\phi}(t, \xi) = \phi(t, \mathcal{L}_t(\xi))$ or $\hat{\mathbf{u}}(t, \xi) = \mathbf{u}(t, \mathcal{L}_t(\xi))$. The flow map $\mathcal{L}_t$ is the change of variables between these descriptions, and the trajectory of a particle with label ξ is the characteristic curve $t \mapsto \mathbf{x}(t; \xi) = \mathcal{L}_t(\xi)$. Where defined, the inverse $\mathcal{L}_t^{-1}$ maps an Eulerian location back to its material label.

Under the standard assumption that the prescribed velocity field is continuous in time and uniformly Lipschitz in space, particle trajectories exist uniquely and depend continuously on their initial labels; see [3, Ch. 2]. This is a trajectory-level ODE result and should not be confused with Navier–Stokes well-posedness.

Definition 1.3 (Material derivative). For $\phi \in C^1(\mathcal{Q}_T)$, the material derivative is the derivative of ϕ along a particle trajectory:

$$D_t \phi(t, \mathbf{x}) := \frac{d}{dt} \phi(t, \mathcal{L}_t(\xi)) = \partial_t \phi(t, \mathbf{x}) + \mathbf{u}(t, \mathbf{x}) \cdot \nabla \phi(t, \mathbf{x}), \quad \xi = \mathcal{L}_t^{-1}(\mathbf{x}).$$

For vector fields, D_t is applied componentwise.

Let $\hat{\mathbf{F}}_t = \nabla_\xi \mathcal{L}_t$ be the deformation gradient and $\hat{J}_t = \det \hat{\mathbf{F}}_t$ the Jacobian determinant. In Eulerian notation this is written $J_t(\mathbf{x}) = \det \hat{\mathbf{F}}_t(\mathcal{L}_t^{-1}(\mathbf{x}))$. The standard Jacobian evolution law $D_t J_t = J_t \operatorname{div} \mathbf{u}$ is used implicitly below; derivation details are deferred to Appendix E.

Theorem 1.4 (Reynolds Transport Theorem). Let $V(t) = \mathcal{L}_t(V(0))$ be a bounded material volume with

Lipschitz boundary. If $\phi \in C^1(\mathcal{Q}_T)$, then

$$\begin{aligned} \frac{d}{dt} \int_{V(t)} \phi \, d\mathbf{x} &= \int_{V(t)} (D_t \phi + \phi \operatorname{div} \mathbf{u}) \, d\mathbf{x} \\ &= \int_{V(t)} (\partial_t \phi + \operatorname{div}(\phi \mathbf{u})) \, d\mathbf{x}. \end{aligned}$$

Proof-level details for Theorem 1.4 and the momentum balance below are found in Appendix E.

Continuum assumptions retained in the main model

Assumption 1.1 supplies the 3D velocity, pressure, density, and stress fields used below. The balance-law vocabulary developed below permits variable density. The reduced arterial models considered later adopt the constant-density specialization $\rho \equiv \rho_0 > 0$, which is standard for large-vessel incompressible blood-flow models.

The domains and material control volumes are assumed regular enough for traces, outward unit normals, and the divergence theorem; Lipschitz boundaries are sufficient for the integral manipulations used here; see Convention D.1 in Appendix D.

Definition 1.5 (Material density preservation). A flow has material density preservation if density is preserved along particle trajectories:

$$D_t \rho = 0.$$

Definition 1.6 (Incompressible flow). A flow is incompressible, or volume preserving, if material volumes are preserved. With $\mathcal{L}_0 = \operatorname{Id}$ this is equivalent to

$$J_t \equiv 1 \quad \Longleftrightarrow \quad \operatorname{div} \mathbf{u} = 0.$$

Remark 1.7 (Divergence-free condition). Mass conservation gives

$$\partial_t \rho + \operatorname{div}(\rho \mathbf{u}) = 0 \quad \Longleftrightarrow \quad D_t \rho + \rho \operatorname{div} \mathbf{u} = 0.$$

In the constant-density model used for incompressible Navier–Stokes, $\rho \equiv \rho_0 > 0$, so mass conservation reduces to $\operatorname{div} \mathbf{u} = 0$ in $\mathcal{Q}_T$.

Definition 1.8 (Linear momentum balance). For a material fluid element $V(t) \subset \Omega_B(t)$, the linear momentum balance is

$$\frac{d}{dt} \int_{V(t)} \rho \mathbf{u} \, dV = \int_{\partial V(t)} \mathbf{T} \hat{\mathbf{n}} \, dS + \int_{V(t)} \rho \mathbf{f}_b \, dV,$$

where $\mathbf{T}$ is the Cauchy stress tensor and $\mathbf{f}_b$ is body force per unit mass.

Definition 1.9 (Strong momentum balance). Under the regularity and mass-balance assumptions above, the local momentum balance is

$$\rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = \operatorname{div}(\mathbf{T}) + \rho \mathbf{f}_b, \quad \text{in } \Omega_B(t).$$

The stress law closes the momentum equation. Let the rate-of-deformation tensor be

$$\mathbf{D}(\mathbf{u}) := \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top).$$

Definition 1.10. A fluid is Newtonian if, after separating the pressure contribution, its viscous stress depends linearly on the rate-of-deformation tensor $\mathbf{D}(\mathbf{u})$.

Remark 1.11 (Blood isotropy justification). Blood is often modeled as an isotropic continuum fluid at large-vessel scales. This is a closure assumption for averaged behavior, not a microscopic statement about cells suspended in plasma. The cited sources support the scale and rheology caveat—nearly constant large-vessel relative viscosity versus diameter- and hematocrit-dependent apparent viscosity in small tubes—not a direct cellular-level isotropy claim [1, Sec. 1.1.3, Fig. 1.14, pp. 12–14, Sec. 3.2, pp. 46–47] [4, Abstract, pp. H1770–H1772].

Remark 1.12 (Isotropic constitutive response). An isotropic fluid has a constitutive response that is independent of the chosen coordinate system. Consequently, the Cauchy stress depends on coordinate-free invariants of $\mathbf{D}(\mathbf{u})$ rather than on a particular basis.

Definition 1.13 (Newtonian, isotropic constitutive law). For a Newtonian, isotropic fluid,

$$\mathbf{T} = -p\mathbf{I} + 2\eta \mathbf{D}(\mathbf{u}) + \lambda \operatorname{div}(\mathbf{u}) \mathbf{I},$$

where η is the dynamic viscosity and λ is the coefficient of the volumetric-viscous term.

Definition 1.14 (Incompressible stress tensor). When $\operatorname{div} \mathbf{u} = 0$, the Newtonian incompressible Cauchy stress is

$$\mathbf{T} = -p\mathbf{I} + 2\eta \mathbf{D}(\mathbf{u}).$$

More general models allow viscosity to vary with shear rate. For generalized Newtonian blood models, the dynamic viscosity is allowed to depend on the local deformation-rate magnitude [2, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19]. A standard constitutive form is

$$\dot{\gamma} := \sqrt{2\mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{u})},$$

$$\mathbf{T} = -p\mathbf{I} + 2\eta_{\text{eff}}(\dot{\gamma})\mathbf{D}(\mathbf{u}).$$

Here

$$\|\mathbf{D}(\mathbf{u})\|^2 := \mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{u}) = \sum_{i,j} D_{ij}(\mathbf{u})^2$$

is the pointwise Frobenius norm squared. When a source writes the viscosity as a function of $D : D$ or $|D|_{\text{F}}^2$, this report translates it through the scalar shear-rate convention above. Here η_{eff} is an effective dynamic viscosity. Kinematic viscosity is $\nu = \eta/\rho$ after division by density. The Newtonian incompressible model is recovered from this generalized Newtonian form when η_{eff} is constant.

Remark 1.15 (Newtonian Blood Justification). For large arteries, empirical studies suggest that apparent viscosity is often approximately constant across the relevant operating range, motivating the Newtonian approximation in many reduced-order hemodynamic models. However, diameter-dependent, hematocrit-dependent, and shear-dependent effects become increasingly important in smaller vessels and low-shear regions [1, Sec. 1.1.3, pp. 12–14] [4].

Together, Assumption 1.1 and the balance-law and stress-closure definitions in this subsection form the continuum contract used by the following hierarchy: material volumes and Reynolds transport support the mass and momentum balances, while incompressibility and the selected constitutive law specialize those balances into the PDE tiers that are later reduced by cross-sectional averaging.

1.2 Governing Equations and Model Hierarchy

The continuum assumptions of Section 1.1 lead to a layered modeling hierarchy. In particular, Assumption 1.1 supplies the flow map, material volumes, and continuum fields on which the conservation laws act. The high-fidelity starting point is the incompressible Navier–Stokes or generalized-Newtonian system on a three-dimensional lumen. The report then uses cross-sectional reduction to identify the selected 1D area-flow variables and to state the closure assumptions that must be fixed before any reduced numerical study is interpreted.

1.2.1 Conservation Laws and Navier–Stokes Specialization

The governing equations originate from conservation of mass and linear momentum applied to material volumes under Assumption 1.1, followed by the constant-density incompressible specialization. The derivation details and row-divergence convention are recorded in Appendix E and Definition D.6. Figure 2 summarizes the route from these balance laws to the forms used in the report [2, Part I, Ch. 1, Secs. 2.2–3.1, pp. 11–19, Part IV, Ch. 1, Sec. 1.1, pp. 380–383] [5, Sec. 6].

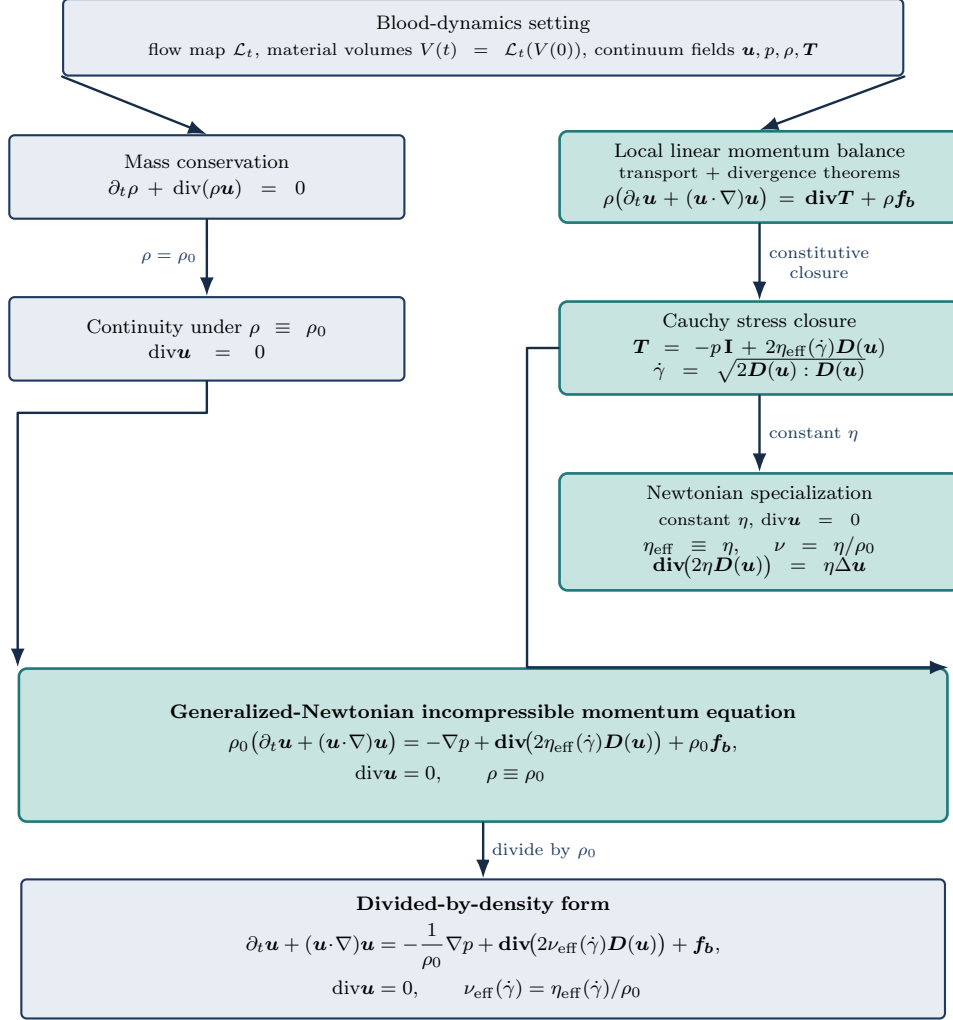


Figure 2: From the blood-dynamics setting to incompressible Navier–Stokes. The flow map and material-volume assumptions support the conservation statements. Mass conservation reduces to $\text{div} \mathbf{u} = 0$ under constant density. Momentum balance closes through the Cauchy stress law; the Newtonian limit uses constant dynamic viscosity, while the generalized-Newtonian form keeps $\eta_{\text{eff}}(\dot{\gamma})$ explicit.

With this kinematic setting fixed, mass conservation for the constant-density incompressible model gives

$$\text{div} \mathbf{u} = 0 \quad \text{in } \Omega_B(t).$$

Linear momentum conservation balances inertia, pressure, viscous stress, and body forces:

$$\rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = \text{div}(\mathbf{T}) + \rho \mathbf{f}_b.$$

Definition 1.16 (Conservative incompressible Navier–Stokes). On $\Omega_B(t)$, the constant-density Newtonian

system may be written in conservative tensor-divergence form as

$$\begin{cases} \partial_t(\rho \mathbf{u}) + \mathbf{div}(\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \mathbf{div}(2\eta \mathbf{D}(\mathbf{u})) + \rho \mathbf{f}, \\ \mathbf{div} \mathbf{u} = 0, \quad \rho \equiv \rho_0 > 0. \end{cases}$$

For constant density and divergence-free velocity, $\mathbf{div}(\rho \mathbf{u} \otimes \mathbf{u}) = \rho(\mathbf{u} \cdot \nabla) \mathbf{u}$. Thus the same Newtonian momentum equation can be read in advective form.

Definition 1.17 (Generalized-Newtonian incompressible Navier–Stokes). For an incompressible generalized-Newtonian fluid, use the shear-rate convention

$$\dot{\gamma} := \sqrt{2\mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{u})}.$$

The velocity $\mathbf{u}$, pressure p , body force $\mathbf{f}$, and constant density $\rho \equiv \rho_0 > 0$ satisfy

$$\begin{cases} \rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = -\nabla p + \mathbf{div}(2\eta_{\text{eff}}(\dot{\gamma}) \mathbf{D}(\mathbf{u})) + \rho \mathbf{f}, \\ \mathbf{div} \mathbf{u} = 0. \end{cases}$$

Definition 1.18 (Density-scaled generalized Navier–Stokes). With $\nu_{\text{eff}}(\dot{\gamma}) := \eta_{\text{eff}}(\dot{\gamma})/\rho$, the divided-by-density form is

$$\begin{cases} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho^{-1} \nabla p + \mathbf{div}(2\nu_{\text{eff}}(\dot{\gamma}) \mathbf{D}(\mathbf{u})) + \mathbf{f}, \\ \mathbf{div} \mathbf{u} = 0. \end{cases}$$

The density-scaled form is particularly useful for numerical simulations, because it separates the convective $((\mathbf{u} \cdot \nabla) \mathbf{u})$ and diffusive $(\mathbf{div}(2\nu_{\text{eff}}(\dot{\gamma}) \mathbf{D}(\mathbf{u})))$ contributions in the momentum equation, allowing for more straightforward numerical treatment of each term independently.

Definition 1.19 (Newtonian incompressible Navier–Stokes). If $\eta_{\text{eff}} \equiv \eta$ is constant, then sufficiently smooth divergence-free velocity fields satisfy $\mathbf{div}(2\eta \mathbf{D}(\mathbf{u})) = \eta \Delta \mathbf{u}$. The Newtonian equations are

$$\begin{cases} \rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = -\nabla p + \eta \Delta \mathbf{u} + \rho \mathbf{f}, \\ \mathbf{div} \mathbf{u} = 0, \end{cases} \quad \begin{cases} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho^{-1} \nabla p + \nu \Delta \mathbf{u} + \mathbf{f}, \\ \mathbf{div} \mathbf{u} = 0, \end{cases}$$

where $\nu := \eta/\rho$.

Thus the 3D PDE tier is not introduced independently: flow maps and material volumes support Reynolds transport, Reynolds transport gives mass and momentum balance, the Cauchy stress law closes momentum, and incompressibility selects the Navier–Stokes systems above.

Definition 1.20 (Navier–Stokes residual notation). The density-scaled generalized system can be summarized by a momentum residual operator:

$$\begin{cases} \mathcal{R}_{\text{NS}}(\partial_t \mathbf{u}, \nabla \mathbf{u}, \nabla p, \mathbf{u}, p; \rho, \nu_{\text{eff}}) = \mathbf{f}, \\ \text{div} \mathbf{u} = 0. \end{cases}$$

Remark 1.21 (Navier–Stokes system classification). Navier–Stokes is a nonlinear coupled PDE system. The unknowns are the three velocity components and pressure. The convective term transports momentum, the pressure gradient enforces incompressibility, and the viscous term diffuses momentum through the stress tensor.

1.2.2 Boundary Data and Well-Posedness Context

For a single-vessel three-dimensional domain, write the boundary decomposition

$$\partial\Omega_B(t) = \Gamma_{\text{in}}(t) \cup \Gamma_{\text{out}}(t) \cup \Gamma_w(t).$$

Here $\Gamma_{\text{in}}(t)$ denotes an inlet portion where a velocity, flow-rate, or profile condition may be prescribed. The outlet portion $\Gamma_{\text{out}}(t)$ carries pressure, traction, or reduced downstream network data. The wall portion $\Gamma_w(t)$ carries either a no-slip or no-penetration condition for a fixed-wall continuum model, or a wall-law or structure closure when wall motion is retained. A complete initial-boundary value problem also states $\mathbf{u}(0, \cdot) = \mathbf{u}_0$, the domain, forcing, compatibility assumptions, and the solution class.

Remark 1.22 (Global regularity context). For smooth, spatially localized initial data in three dimensions, global existence and smoothness for Navier–Stokes remains an open Clay Millennium Prize problem [6, Problem statement]. This is mathematical context for the 3D continuum model, not a direct limitation of the controlled reduced 1D solver introduced below.

Classical local well-posedness statements require a domain, boundary conditions, forcing, initial-data space, and solution class. Under compatible smooth data one obtains a unique strong solution on a maximal time interval. The domain and function-space notation used in such statements is summarized in Appendix D; coordinate and energy background is collected in Appendix F.

1.2.3 Cross-Sectional Reduction and Selected 1D Model

The 1D model reduces the spatial dimension by assuming that the vessel has a distinguished axial direction and that pressure and velocity can be averaged over cross-sections. This keeps pulse-wave and pressure-drop structure while discarding secondary flow, separation, recirculation, and resolved wall traction. The

reduction follows standard compliant-artery and dimension-reduced modeling arguments [7, Sec. 2, pp. 253–257] [1, Ch. 3, Secs. 3.2 and 3.6] [8, Sec. 3, pp. 24–26]. The reduced variables A , Q , $\bar{u}$, and p_g should therefore be read as cross-sectional outputs of the continuum fields under additional averaging, geometry, wall-law, and profile assumptions, not as independent replacements for the blood-dynamics setting above.

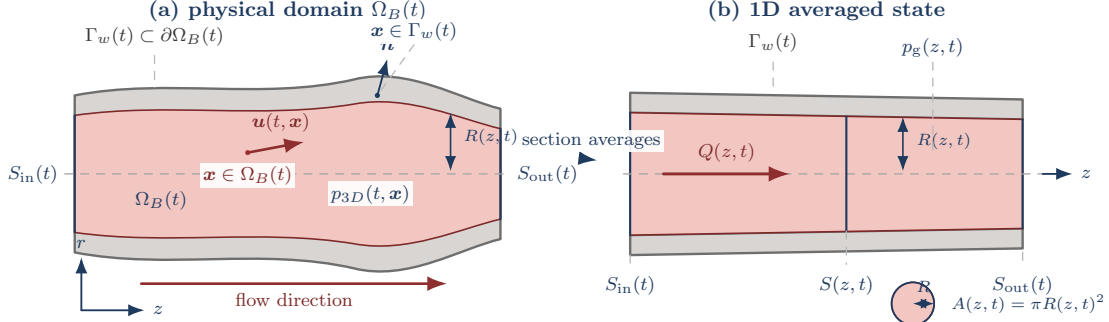


Figure 3: Compliant-vessel notation for the 1D reduction. The physical lumen $\Omega_B(t)$ is represented by a straightened averaging domain with cross-section $S(z, t)$, area $A(z, t)$, flow $Q(z, t)$, radius $R(z, t)$, and mean gauge pressure $p_g(z, t)$.

Assumption 1.23 (Straight single-vessel geometry). The selected 1D lane assumes a vessel of length L aligned with the z -axis and a circular cross-section with radius $R(z, t)$. For $0 < z < L$, $S(z, t)$ is the interior cross-section used for averaging, not a boundary component of the open lumen.

Definition 1.24 (Section-averaged 1D fields). The retained reduced variables are

$$\begin{aligned}
 A(z, t) &:= \int_{S(z, t)} 1 \, dS, && \text{cross-sectional area,} \\
 Q(z, t) &:= \int_{S(z, t)} u_z \, dS, && \text{volumetric flow rate,} \\
 \bar{u}(z, t) &:= \frac{Q(z, t)}{A(z, t)}, && \text{mean axial velocity,} \\
 p_g(z, t) &:= \frac{1}{A(z, t)} \int_{S(z, t)} p_{3D, g} \, dS, && \text{mean gauge pressure.}
 \end{aligned}$$

Here $p_{3D, g}$ is continuum pressure in the wall-law gauge frame. The pressure-reference convention for reduced ratios is fixed separately in Convention C.2.

Definition 1.25 (Closed 1D area-flow model). For an incompressible fluid with density ρ , the working compliant single-vessel model is

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial z} = 0, \quad (1.1)$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial z} \left(\alpha \frac{Q^2}{A} \right) + \frac{A}{\rho} \frac{\partial p_g(A, z, t)}{\partial z} = -C_f(z, A, Q) \frac{Q}{A}. \quad (1.2)$$

The parameter α is the momentum-flux correction factor, C_f is the wall-friction closure, and $p_g(A, z, t)$ is the wall-law gauge pressure closing the reduced system [7, Sec. 2, pp. 253–255] [1, Ch. 3, Sec. 3.2, pp. 40–47].

Assumption 1.26 (First-run momentum-profile and friction closure). The baseline reduced numerical lane fixes $\alpha = 1$ and uses a compact Poiseuille-calibrated wall-friction coefficient

$$C_f := \frac{8\pi\eta}{\rho}.$$

This is a numerical contract for the first controlled 1D study. It should not be read as a claim that the flat-profile flux convention, friction law, and any derived shear proxy form a complete resolved velocity-profile model.

Definition 1.27 (Elastic pressure-area wall law). The baseline pressure closure is the elastic wall law

$$p_g(A, z, t) = p_{\text{ext}}(z, t) + \frac{\beta(z)}{A_0(z)} \left(\sqrt{A(z, t)} - \sqrt{A_0(z)} \right). \quad (1.3)$$

Here $A_0(z)$ is the reference area, p_{ext} is the external pressure, and $\beta(z)$ is a wall-stiffness parameter. Some implementations absorb A_0^{-1} into β ; the essential closure is that p_g is known once A_0 , β , and p_{ext} are supplied.

Definition 1.28 (Stenosis reference geometry). A stenosis is encoded through the reference radius or reference area. Let $R_{\text{base}}(z)$ be the nominal healthy radius and $s(z) \in [0, 1)$ a fractional radius-reduction profile. Then

$$R_0(z) = R_{\text{base}}(z)(1 - s(z)), \quad A_0(z) = \pi R_0(z)^2. \quad (1.4)$$

The baseline idealized stenotic-vessel profile is the smooth asymmetric kernel

$$g(z) = z - z_s + \kappa \exp \left(-\frac{(z - z_a)^2}{2\sigma_a^2} \right), \quad \psi(z) = \exp \left(-\lambda g(z)^4 \right), \quad (1.5)$$

$$s(z) = s_{\text{max}} \psi(z), \quad 0 \leq s_{\text{max}} < 1. \quad (1.6)$$

In the default centimeter-scale simulation record, $z_a = 2.5$ cm, $z_s = 3.4$ cm, $\sigma_a = 1$ cm, $\kappa = 0.95$ cm, and $\lambda = 50 \text{ cm}^{-4}$. The geometry is an idealized numerical design choice, not a patient-specific plaque morphology or clinical calibration.

Remark 1.29 (Stenosis-profile and variable-radius boundary). The profile in Eq. 1.6 is C^∞ and rapidly localized rather than compactly supported. This smooth idealized geometry is the baseline for the reduced simulations in this report. Canic et al. show that standard 1D variable-geometry models may miss stenotic pressure and velocity trends unless additional variable-radius terms are included [9, Abstract, Secs. 1–2]. Those corrections motivate later comparison questions, but they are not active in the baseline model unless a separate closure identifier and numerical contract are declared.

Definition 1.30 (Conservative flux and source notation). The pressure term in Eq. 1.2 can be represented

through an elastic-potential flux plus explicit sources:

$$\partial_t A + \partial_z Q = 0, \quad (1.7)$$

$$\partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} + \Psi(A, z, t) \right) = S_{\text{geom}}(A, z, t) + S_f(A, Q, z), \quad (1.8)$$

where

$$\begin{aligned} \partial_A \Psi(A, z, t) &= \frac{A}{\rho} \partial_A p_g(A, z, t), \\ S_f(A, Q, z) &= -C_f(z, A, Q) \frac{Q}{A}. \end{aligned}$$

This notation records the balance-law objects needed by finite-volume schemes; it does not assert that every later implementation treats the flux and source terms identically [10] [11, Summary, Secs. 2.1–2.3, Secs. 5–6].

Definition 1.31 (Single-vessel boundary data). For a single-vessel solve on $z \in [0, L]$, prescribe one inlet condition and one outlet condition:

$$\begin{aligned} Q(0, t) &= Q_{\text{in}}(t) \quad \text{or} \quad \bar{u}(0, t) = \bar{u}_{\text{in}}(t), \\ p_g(L, t) &= p_{\text{out},g}(t) \quad \text{or} \quad p_g(L, t) = p_{\text{WK},g}(Q(L, \cdot), t). \end{aligned}$$

The second outlet option denotes a terminal Windkessel-style relation. The first controlled study uses prescribed $Q_{\text{in}}(t)$ and $p_{\text{out},g}(t)$ because they give an unambiguous reduced single-vessel contract.

Assumption 1.32 (Subcritical boundary regime). The baseline boundary treatment is used only in the subcritical regime $\lambda_- < 0 < \lambda_+$, where one characteristic enters through the inlet and one characteristic enters through the outlet.

Under this regime, a finite-volume boundary realization constrains the incoming information at each boundary and leaves the outgoing state determined by the interior solve. This is the reduced-model analogue of declaring enough boundary data for the selected equation class; it is not, by itself, a validation of a particular boundary discretization.

1.2.4 Model Hierarchy and Comparison Boundary

Reduced hemodynamic models form a hierarchy of modeling roles rather than a simple accuracy ranking. A higher-dimensional model becomes a comparison only after the kinematic and continuum assumption package, geometry, wall response, rheology, inlet and outlet data, numerical resolution, output operators, and discrepancy metrics are matched.

Across these categories, the recurring descriptors are the kinematic/continuum setting, stenosis geometry,

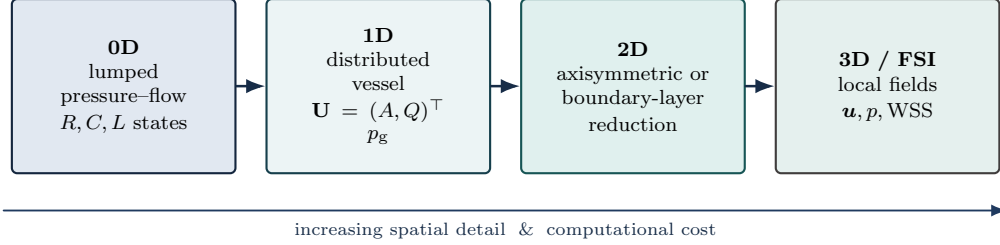


Figure 4: Model hierarchy for reading comparison claims. The present manuscript selects a 1D reduced forward model; 3D, FSI, clinical, and independent-solver comparisons require separately matched evidence.

Table 1: Hemodynamic model taxonomy for the idealized stenosis report.

Model family	Typical role	Common states or outputs	Report boundary
0D lumped circulation	Outlet loads, Windkessel context, and network hemodynamic summaries	Compartment pressures, flows, resistances, compliances, inertances	Useful for boundary conditions, not for local stenosis geometry or resolved wall traction
Classical 1D compliant-vessel models	Distributed area-flow description after cross-sectional averaging	$A(z, t)$, $Q(z, t)$, $p_g(z, t)$, wave speed, pressure drop	Main reduced model tier under declared continuum, wall, friction, inlet, and outlet assumptions
Stenosis-aware 1D and pressure-loss ROMs	Idealized stenosis trends, variable-radius corrections, or algebraic loss estimates	Stenosis profile, severity, flow rate, Δp , pressure ratio, reduced shear proxy	Reduced-order trend context only unless matched against resolved or clinical evidence
2D, 3D, and FSI models	Higher-fidelity local flow and wall-motion comparison	Velocity, pressure, wall traction, WSS, wall displacement, mesh and time diagnostics	Reference-tier context unless the same continuum setting, geometry, rheology, wall, boundary data, and outputs are compared directly
Numerical reference problems for 1D solvers	Scheme comparison, reproducibility expectations, and implementation context	Grid, timestep, flux and source choices, boundary states, field and scalar discrepancies	Evidence standard; does not grant solver parity to the present code without a matched comparison

wall stiffness or compliance, density, viscosity or effective-viscosity law, inlet waveform, outlet pressure or load, pressure-reference convention, and tap locations. The recurring outputs are pressure drop, reduced pressure ratio, area, flow, radius, and either a reduced wall-friction/shear proxy or a resolved traction-based WSS quantity [1, 7, 8] [9, 12]. The report therefore treats the selected 1D area-flow model as the controlled forward map for the first numerical study, while reserving 3D/FSI agreement, clinical calibration, and independent-solver parity for later evidence.

1.3 Clinical Pressure–Flow Motivation

Anatomy–function distinction. Coronary stenosis is the motivating lesion class because a visible narrowing does not automatically imply a flow-limiting obstruction. Clinical assessment therefore distinguishes anatomical stenosis detection from functional pressure–flow assessment [13, p. 4401] [14, Ch. 1, p. 15, Fig. 1.14, Ch. 2, Secs. 2.1–2.2, pp. 22–23, Ch. 4, Secs. 4.1 and 4.3, pp. 56–57] [15, Sec. 1, Introduction, pp. 390–391]. Coronary lesion-imaging studies likewise show that anatomy-side lesion description carries local geometry and length context, but those descriptors still remain distinct from the functional pressure–flow outputs treated here [16, Abstract, Methods, Table 2, Tables 3–5]. This anatomy–function split motivates

reporting pressure drop, reduced pressure ratio, and reduced shear information rather than treating stenosis severity $s_{\max}$ as the output. Figure 5 summarizes this motivation as a schematic distinction between visible narrowing and pressure–flow loss.

Clinical fractional flow reserve (FFR) is an invasive pressure-wire index, conventionally interpreted in its measurement setting as distal coronary pressure divided by aortic pressure during maximal hyperemia [14, Ch. 5, Sec. 5.5, pp. 72–73] [17, Abstract, Background]. FAME lesion-level evidence documented the discordance between angiographic and functional severity and is the clinical source for the familiar $\text{FFR} \leq 0.80$ decision context. [18, Abstract, Results and Conclusions]. Computed tomography-derived FFR (CT-FFR) sources supply noninvasive comparison vocabulary within their reported study designs: they compare pressure–flow quantities inferred from coronary computed tomography angiography (CCTA)-derived anatomy against invasive FFR or related reference measurements [19, Abstract, Methods, Results and Conclusions] [15, Abstract, Sec. 1, Introduction, Secs. 1.1–1.2, pp. 390–392].

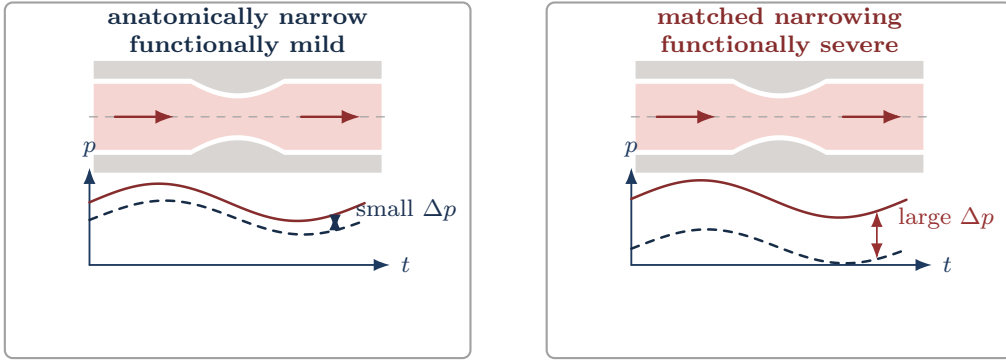


Figure 5: Pressure–flow motivation for the reduced-output contract. The schematic distinguishes an anatomical narrowing from the pressure-loss signature used by reduced-model outputs.

For the selected model pressure field, define the observed proximal and distal pressure traces by

$$P_{\text{prox}}(t) := \mathcal{P}_{\text{prox}}[p(\cdot, t)], \quad P_{\text{dist}}(t) := \mathcal{P}_{\text{dist}}[p(\cdot, t)].$$

The pressure-drop waveform is the additive-datum-invariant quantity

$$\Delta p(t) := P_{\text{prox}}(t) - P_{\text{dist}}(t).$$

Unlike $\Delta p(t)$, a pressure ratio is not invariant under additive pressure shifts, so the ratio convention must declare a common additive pressure reference. The manuscript’s convention adds that reference after the pressure observations have been taken. With the declared p_{ref} , define

$$P_{\text{prox}}^{\text{ratio}}(t) := p_{\text{ref}} + P_{\text{prox}}(t), \quad P_{\text{dist}}^{\text{ratio}}(t) := p_{\text{ref}} + P_{\text{dist}}(t).$$

If a selected observation functional preserves constant shifts, then these ratio-pressure traces agree with

observing $p_{\text{ref}} + p(\cdot, t)$ directly. The definition above does not require that property; it fixes the ratio convention at the level of observed scalar traces. For an admissible ratio-output specification whose observed proximal and distal ratio-pressure traces are integrable on the window and whose declared denominator rule makes the proximal averaged denominator well defined and nonzero, the reduced pressure ratio is

$$\text{PR}_{1D}(\pi_{\text{PR}}) := \frac{\langle P_{\text{dist}}^{\text{ratio}}(t) \rangle_{\mathcal{T}_{\text{HF}}}}{\langle P_{\text{prox}}^{\text{ratio}}(t) \rangle_{\mathcal{T}_{\text{HF}}}}.$$

The 1D specialization used later is the case in which $\mathcal{P}_{\text{prox}}$ and $\mathcal{P}_{\text{dist}}$ are evaluated from the declared axial pressure taps or cross-sectional pressure averages; see Figure 6.

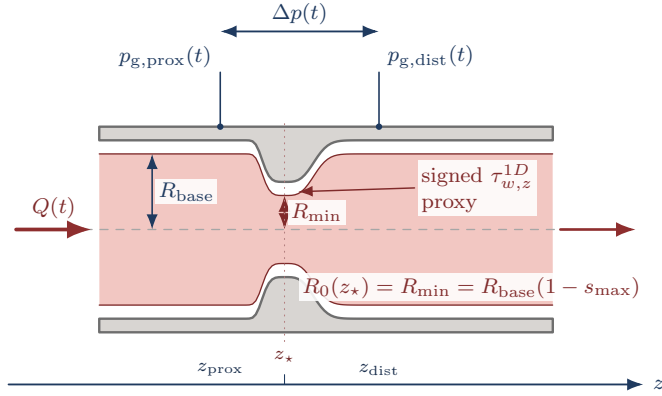


Figure 6: Schematic tap and output convention for an idealized stenosis, not drawn from a computed case. The taps define $\Delta p(t)$; π_{PR} additionally declares the averaging window, pressure reference, and denominator rule needed for $\text{PR}_{1D}(\pi_{\text{PR}})$.

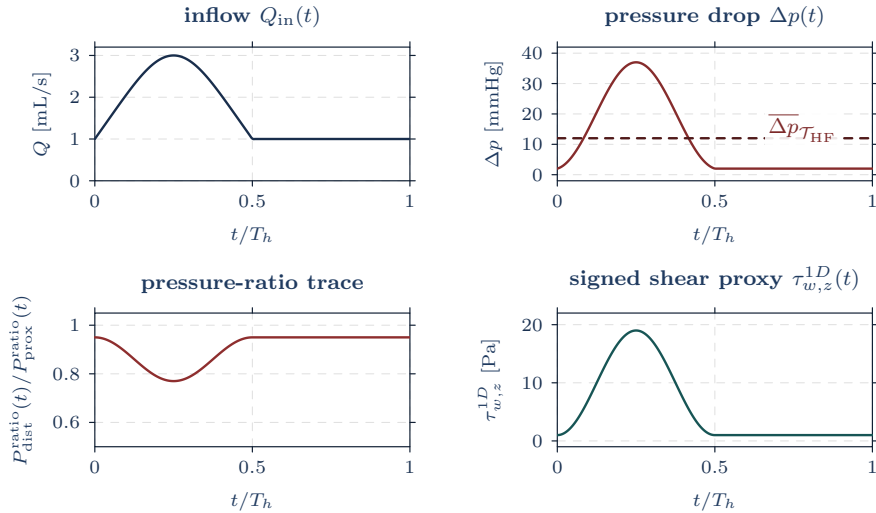


Figure 7: Reduced-output signals over one schematic cardiac cycle. The imposed inflow trace, pressure-drop waveform, metadata-dependent pressure ratio, and signed 1D shear proxy illustrate the declared output semantics, not simulated data or acceptance evidence.

Resolved wall shear stress (WSS) and reduced shear proxy. Pressure ratios and pressure drops summarize lesion-scale functional loss in the selected model tier. Wall shear stress describes local wall loading in a resolved state. These roles are related but not interchangeable. When the stress trace is well defined, resolved WSS is the tangential part of the wall traction obtained from a resolved stress tensor and wall normal; a signed component additionally requires a chosen wall-tangent direction [20, Abstract, Sec. 2, pp. 8–10]. The reduced shear proxy used later is therefore a tier-specific closure diagnostic, separate from resolved traction-based WSS.

2 Resolved 3D Comparison at $T = 1$

This section records the numerical comparison produced by the Julia implementation of the stenosis-aware 1D area-flow model against the available resolved 3D velocity data. The comparison is an output-operator check, not a clinical validation claim and not an exact cross-section quadrature of the 3D tetrahedral solution. The 3D diagnostic averages node-centered axial velocity over thin axial slabs and compares those values with the interpolated 1D quantity Q/A . Radial profiles use the same slab nodes binned by $\sqrt{x^2 + y^2}/R_0(z)$ and compare them with the 1D closure profile.

The two included upstream states are case 77, treated as a 23% stenosis, and case 60, treated as a 40% stenosis. Both XDMF files report $t = 0.999499999999453$, within 5.0×10^{-4} of $T = 1$. Upstream case 50 was not used because its XDMF time is $t = 1.499499999998904$, outside the 10^{-3} tolerance for the $T = 1$ comparison.

Table 2: Section-mean and radial-profile discrepancy summary for the $T = 1$ resolved-node comparison. Velocity errors are in cm/s.

Case	Severity	Sections	Min nodes	Mean $ e_s $	Max $ e_s $	Mean rel.	Max rel.	Mean $ e_r $	Max $ e_r $
77	23%	200	64	7.471	26.26	0.4534	1.622	10.03	23.01
60	40%	200	43	9.440	47.48	0.5141	2.458	15.35	41.72

Figure 8 shows the section-mean comparison. The 1D model captures the smooth upstream and downstream trend but overpredicts the highest-velocity region near the stenosis, especially for the 40% case. The largest section errors occur for case 60 around $z = 2.4$ – 2.8 cm, where the 1D mean velocity is between 58 and 67 cm/s while the node-slab 3D mean is between 19 and 27 cm/s.

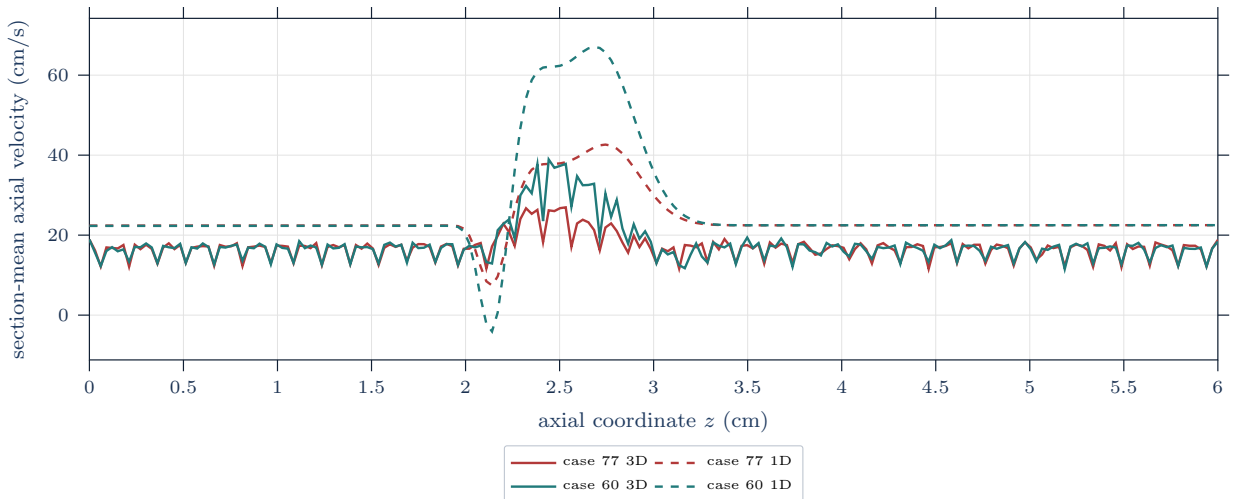


Figure 8: Section-mean axial velocity comparison at $T = 1$. Solid curves are resolved 3D node-slab means; dashed curves are interpolated 1D Q/A .

The radial-profile diagnostics in Figure 9 show the same pattern at the profile level. For both cases, the throat slice $z = 2.451$ cm has fewer populated radial bins because fewer 3D nodes lie in the narrowest cross-

Table 3: Five largest section-mean absolute errors across the two included $T = 1$ cases. Velocities and errors are in cm/s.

Rank	Case	z (cm)	u_{1D}	u_{3D}	$ e $	Rel.	Nodes
1	60	2.714	66.79	19.31	47.48	2.458	72
2	60	2.774	63.74	24.57	39.17	1.594	60
3	60	2.412	61.90	23.51	38.39	1.633	67
4	60	2.563	63.71	27.15	36.56	1.347	72
5	60	2.834	57.59	21.68	35.92	1.657	82

section slab. The largest radial-profile discrepancy is downstream of the throat at $z = 2.951$ cm, where case 60 reaches a maximum per-bin absolute error of 41.72 cm/s.

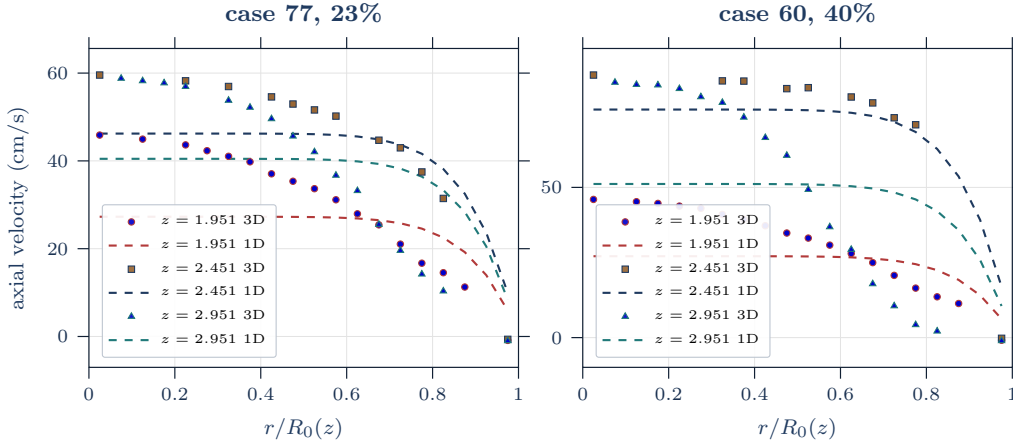


Figure 9: Radial velocity profile comparison at the default throat-centered slices. Markers are resolved 3D radial-bin means; dashed curves are the 1D closure profile reconstructed from the local Q/A .

Table 4: Radial-profile discrepancy by case and slice. Errors are in cm/s.

Case	z (cm)	Populated bins	Mean $ e_r $	Max $ e_r $
77	1.951	17	9.404	18.60
77	2.451	12	6.949	13.33
77	2.951	16	13.01	23.01
60	1.951	18	10.04	18.90
60	2.451	10	7.562	17.17
60	2.951	17	25.55	41.72

These discrepancies should be read in light of the comparison boundary in Section 1.2.4. The present 1D solver is compared to a resolved 3D velocity field through simple node-slab output operators. The result is useful for locating where the reduced velocity closure departs from the resolved state, but it does not by itself identify whether the discrepancy comes from the 1D closure, the boundary data, the 3D post-processing operator, or differences between the numerical formulations.

A Acronyms and Abbreviations

General abbreviations.

a.e. almost everywhere bpm beats per minute
e.g. *exempli gratia* (for example) i.e. *id est* (that is)

Clinical, modeling, and numerical acronyms.

CAS	coronary artery stenosis	FSI	fluid–structure interaction
CCTA	coronary computed tomography angiography	RBC	red blood cell
CT-FFR	computed tomography-derived fractional flow reserve	FFR	fractional flow reserve
IVUS	intravascular ultrasound	WSS	wall shear stress
OCT	optical coherence tomography	CFD	computational fluid dynamics
ODE	ordinary differential equation	PDE	partial differential equation
IC	initial condition	BC	boundary condition
FDM	finite difference method	FEM	finite element method
FVM	finite volume method	ROM	reduced-order model
0D	zero-dimensional	1D	one-dimensional
2D	two-dimensional	3D	three-dimensional

B Mathematical Notation

Notation B.1 (Report mathematical notation). This appendix collects the mathematical notation used throughout this report.

Symbol	Meaning
$:=$	defines
$\equiv$	notational equivalence / identified with
$\mathbb{R}$	real numbers
$\mathbb{R}^+$	strictly positive real numbers
$\mathbb{R}^-$	strictly negative real numbers
$\mathbb{R}^n$	n-dimensional real vector space
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	nonnegative integers $\{0, 1, 2, \dots\}$
$\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$	general basis of $\mathbb{R}^n$
$\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$	standard basis of $\mathbb{R}^n$
$[n]$	index set $\{1, 2, \dots, n\} \subset \mathbb{N}$ for $n \in \mathbb{N}$
I_T	open time interval $(0, T_{\text{final}})$ for transient interior-time arguments
$\bar{I}_T$	closed time interval $[0, T_{\text{final}}]$ for initial-time data and flow maps
K	compact space-time rectangle, such as $[0, \ell] \times [0, T_{\text{final}}]$, used for output-field norms when locally stated
$\Omega \subset \mathbb{R}^n$	bounded domain
$\bar{\Omega}$	the closure of Ω
$\partial\Omega$	the boundary of Ω
$\Omega_B(t)$	spatial blood domain at time t
$\mathcal{Q}_T$	space-time fluid region $\{(t, \mathbf{x}) : t \in I_T, \mathbf{x} \in \Omega_B(t)\}$
$C^k(\Omega)$	functions with continuous classical derivatives through order k in Ω
$C^k(\bar{\Omega})$	functions whose derivatives through order k extend continuously to $\bar{\Omega}$
$C_c^k(\Omega)$	k times continuously differentiable functions with compact support in Ω
$C_c^\infty(\Omega)$	smooth functions with compact support in Ω

Symbol	Meaning
$C(K)$	continuous functions on a stated compact set K
$L^p(\Omega)$	Lebesgue space of p -integrable functions modulo equality a.e.
$d\mathbf{x}$	Lebesgue volume element on $\mathbb{R}^n$
$dS_{\mathbf{x}}$	surface measure element on $\partial\Omega \subset \mathbb{R}^n$
dV	3D volume element on domain $\Omega \subset \mathbb{R}^3$
dS	3D surface element on boundary $\partial\Omega \subset \mathbb{R}^3$
∇	gradient operator
$\Delta\phi = \nabla \cdot \nabla\phi$	Laplacian of scalar field ϕ
div	divergence of a vector field
$\mathbf{div}$	row-wise divergence of a tensor field
v_i	i -th component of vector $\mathbf{v}$
$\mathbf{S} : \mathbf{T}$	Frobenius contraction $\sum_{i,j} S_{ij}T_{ij}$ for second-order tensors
$\langle \cdot, \cdot \rangle_X$	inner product on vector space X
$\langle \mathbf{u}, \mathbf{v} \rangle \equiv \langle \mathbf{u}, \mathbf{v} \rangle_{\mathbb{R}^n}$	inner product of vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$
$\partial_{\hat{\mathbf{n}}}\phi$	normal derivative $\langle \nabla\phi, \hat{\mathbf{n}} \rangle$ of scalar field ϕ on $\partial\Omega$
$\ \cdot\ _X$	norm on a specified normed space X ; bare $\ \cdot\ $ is local shorthand
$\ \cdot\ _F$	Frobenius norm of a matrix or second-order tensor

C Domain Specific Notation

Definition C.1 (SI unit convention). Unless otherwise stated, physical units are SI: length in meters, time in seconds, density in kg/m^3 , pressure in Pa, dynamic viscosity in $\text{Pa} \cdot \text{s}$, and kinematic viscosity in m^2/s .

Convention C.2 (Pressure-reference and ratio convention). Pressure-ratio outputs are reported only with a recorded pressure reference p_{ref} and a gauge pressure p_g in the same reference frame. The reference-consistent pressure used in ratios is

$$p_{\text{ratio}} := p_{\text{ref}} + p_g.$$

The ratio-output specification is

$$\pi_{\text{PR}} := (z_{\text{prox}}, z_{\text{dist}}, \mathcal{T}_{\text{HF}}, p_{\text{ref}}).$$

The subscript HF labels a declared high-flow averaging window in the study record; it is not a clinical hyperemia model unless an external measurement comparison is separately specified [14, Ch. 5, Sec. 5.5, pp. 72–73] [17, p. 1703]. The ratio-output specification is admissible for a computed pressure trace when $\mathcal{T}_{\text{HF}} \subset I_T$ is Lebesgue-measurable with $0 < |\mathcal{T}_{\text{HF}}| < \infty$, the proximal and distal p_{ratio} traces are integrable on that window, and

$$\langle p_{\text{ratio}}(z_{\text{prox}}, t) \rangle_{\mathcal{T}_{\text{HF}}} \neq 0.$$

For an admissible π_{PR} , the corresponding nonclinical pressure ratio is

$$\text{PR}_{1D}(\pi_{\text{PR}}) := \frac{\langle p_{\text{ratio}}(z_{\text{dist}}, t) \rangle_{\mathcal{T}_{\text{HF}}}}{\langle p_{\text{ratio}}(z_{\text{prox}}, t) \rangle_{\mathcal{T}_{\text{HF}}}}.$$

For an integrable scalar trace,

$$\langle f \rangle_{\mathcal{T}_{\text{HF}}} := \frac{1}{|\mathcal{T}_{\text{HF}}|} \int_{\mathcal{T}_{\text{HF}}} f(t) dt.$$

This bracket denotes time averaging, not the inner product notation in Appendix B. An FFR-like shorthand refers to this reduced ratio only for records carrying an admissible π_{PR} ; the shorthand does not add a clinical measurement model [14, Ch. 5, Sec. 5.5, pp. 72–73] [17, p. 1703]. If π_{PR} is absent or inadmissible, the pressure ratio is outside the reported output space.

Symbol	Meaning	Units
R	vessel radius; vessel diameter is $d_v := 2R$	m
d_v	vessel diameter	m
η	dynamic viscosity	$\text{Pa} \cdot \text{s}$
$\nu := \eta/\rho$	kinematic viscosity	m^2/s
$\eta_{\text{eff}}(\dot{\gamma})$	effective dynamic viscosity for generalized Newtonian models	$\text{Pa} \cdot \text{s}$
τ	shear stress	Pa
$\dot{\gamma}$	shear rate	s^{-1}
ρ	density field	kg/m^3
p	pressure field	Pa
$\mathbf{u}$	velocity field	m/s
$D(\mathbf{u})$	rate-of-deformation tensor, $D(\mathbf{u}) := \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$	s^{-1}
$W_0 := d_v \sqrt{\frac{\omega \rho}{\eta}}$	Womersley number, diameter-based convention	—
Re	Reynolds number	—
Pe	Péclet number	—
c_{conc}	concentration field when a transport scalar is discussed	—
D_m	molecular diffusion coefficient	m^2/s
t	time	s
T or T_{final}	terminal time; use T_{final} where it could be confused with stress tensors	s
ω	angular frequency	rad/s
(r, θ, z)	cylindrical coordinates used only when 3D coordinate forms are written	m, —, m
$(\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z)$	cylindrical orthonormal basis vectors	—
τ_{ij}	cylindrical viscous-stress component	Pa
$\mathbf{f}_b$	body force per unit mass	m/s^2
$\rho \mathbf{f}_b$	body force per unit volume	N/m^3

Reduced-model and forward-map notation.

Symbol	Meaning	Units
$A(z, t)$	cross-sectional lumen area in the 1D model	m^2
$Q(z, t)$	volumetric flow rate through a cross-section	m^3/s
$\bar{u}(z, t) := Q/A$	section-mean axial velocity	m/s
$I_z = (0, L)$	axial single-vessel interval for a 1D formulation	m
$\mathbf{U} = (A, Q)^\top$	conservative area-flow state vector	$(\text{m}^2, \text{m}^3/\text{s})$
α	prescribed momentum-flux coefficient in the 1D balance-law closure; the baseline flat-profile convention is $\alpha = 1$	—
$R(A(z, t)) = \sqrt{A(z, t)/\pi}$	current reduced lumen radius derived from the 1D area	m
$R_0(z)$	reference stenosed radius used by the wall law	m
$R_{\text{base}}(z)$	nominal healthy baseline radius before applying $s(z)$	m
$\mathcal{R}$	rheology descriptor in a_{1D} ; distinct from radius symbols R , R_0 , and R_{base}	mixed
$s(z)$	dimensionless stenosis profile, with $0 \leq s(z) < 1$	—
$A_0(z) = \pi R_0(z)^2$	wall-law reference area after applying the stenosis profile	m^2
s_{max}	maximum fractional radius-reduction parameter in $s(z)$	—
$g(z)$	smooth asymmetric stenosis-kernel coordinate	m
$\psi(z) = \exp(-\lambda g(z)^4)$	smooth rapidly localized stenosis kernel used by the baseline profile $s(z) = s_{\text{max}}\psi(z)$	—
z_a	center of the Gaussian asymmetry term in $g(z)$	m
z_s	axial shift parameter in $g(z)$	m
σ_a	width parameter for the Gaussian asymmetry term in $g(z)$	m
κ	amplitude of the Gaussian asymmetry term in $g(z)$	m
λ	localization strength in $\psi(z) = \exp(-\lambda g(z)^4)$	m^{-4}
$\beta(z)$	wall-stiffness parameter in $p_g = p_{\text{ext}} + \beta A_0^{-1}(\sqrt{A} - \sqrt{A_0})$	$\text{Pa} \cdot \text{m}$
$p_{\text{ext}}(z, t)$	external-pressure datum in the wall-law gauge convention; baseline value $p_{\text{ext}} \equiv 0$ unless declared otherwise	Pa
$\Psi(A, z, t)$	elastic flux potential satisfying $\partial_A \Psi = (A/\rho)\partial_A p_g$ and normalized by $\Psi(A_0(z), z, t) = 0$	m^4/s^2
$\mathbf{F}$	area-flow flux map; not a residual or objective functional	problem-dependent

Symbol	Meaning	Units
$\mathbf{S}$	source-term map in the 1D balance-law template	problem-dependent
$\hat{\mathbf{F}}_{i+1/2}^n$	numerical flux at a finite-volume cell interface and time level	problem-dependent
$\hat{\mathbf{S}}_i^n$	cell source quadrature used by a computed finite-volume scheme	problem-dependent
$\Delta z_i, \Delta t^n$	finite-volume cell length and timestep in a computed solve	m, s
C_f or $C_f(z, A)$	wall-friction coefficient in the 1D source $-C_f Q/A$; first-run value $C_f = 8\pi\eta/\rho$	m ² /s
$\eta_{\text{fric}}(z, t)$	dynamic viscosity used by the reduced wall-friction and signed-shear-proxy formula; $\eta_{\text{fric}} = \eta$ for constant-viscosity records and $\eta_{\text{fric}} = \eta_{\text{eff}}(\dot{\gamma}_w)$ for the admitted Carreau–Yasuda wall-friction extension	Pa · s
c_{wave} or c	1D pulse-wave speed, $c^2 = (A/\rho)\partial_A p_g$ when no concentration field is in scope	m/s
$e \in \mathcal{E}$	expert/closure record for a closure-indexed 1D forward-map family; suppress e only for single-expert baseline records	mixed
$\mathcal{G}_{1D,h}^{r_h,e}$	computed fixed-grid 1D map indexed by numerical record r_h and expert record e	mixed

Symbol	Meaning	Units
p_{3D}	continuum pressure in the 3D/NS setting	Pa
$p_{3D,g}$	continuum pressure after subtracting the wall-law gauge reference	Pa
p_g	1D gauge pressure produced by the wall law	Pa
$p_{\text{ratio}} := p_{\text{ref}} + p_g$	reference-consistent pressure used in ratios by Convention C.2	Pa
p_{ref}	fixed pressure reference recorded by the ratio-output specification	Pa
$\pi_{\Delta p} = (z_{\text{prox}}, z_{\text{dist}})$	pressure-drop tap specification for the base 1D output map	m, m
π_{PR}	ratio-output specification $(z_{\text{prox}}, z_{\text{dist}}, \mathcal{T}_{\text{HF}}, p_{\text{ref}})$	mixed
$\langle f \rangle_{\mathcal{T}_{\text{HF}}}$	time average of an integrable trace over the declared high-flow window; not the Appendix B inner product	same as f
$p_{\text{out},g}$	prescribed outlet gauge pressure in the same reference frame as p_g	Pa
$\Delta p(t)$	proximal-minus-distal pressure drop computed from p_g	Pa
$\overline{\Delta p}_{\mathcal{T}_{\text{HF}}}$	scalar pressure-drop output averaged over recorded high-flow window $\mathcal{T}_{\text{HF}}$	Pa
FFR-like shorthand	optional shorthand for PR_{1D} only when Convention C.2 and admissibility requirements are met	—
$\text{PR}_{1D}(\pi_{\text{PR}})$	schematic nonclinical pressure-ratio quantity under the same convention	—
τ_w or $ \tau_w $	resolved traction-based WSS magnitude in 3D contexts	Pa
$\tau_{w,z}^{1D}$	signed 1D axial shear proxy from Q , A , and the recorded wall-friction/rheology closure	Pa

Remark C.3 (Viscosity notation). We write η for dynamic viscosity and

$$\nu := \frac{\eta}{\rho}$$

for kinematic viscosity. The distinction is dimensional: this report follows the convention that the coefficient in the Cauchy stress tensor is a dynamic viscosity: $D(\mathbf{u})$ has units of inverse time, so $\eta D(\mathbf{u})$ has pressure units. For generalized Newtonian models, the scalar viscosity function multiplying $2D(\mathbf{u})$ in the extra stress is likewise an effective dynamic viscosity, denoted

$$\eta_{\text{eff}}(\dot{\gamma}), \quad \dot{\gamma} := \sqrt{2D(\mathbf{u}) : D(\mathbf{u})}.$$

Here “this quantity” means the stress-level viscosity coefficient, not the divided-by-density coefficient. Several sources write the generalized viscosity function as $\mu(|D|^2)$, while related hemodynamics sources also use μ for viscosity coefficients in reduced and continuum settings [1, Sec. 3.1, p. 35, Sec. 3.6, p. 215] [20, Sec. 1.1, pp. 6–7, Eqs. (1.2)–(1.4), Fig. 4.4, p. 18]. To keep this report unambiguous, ν is always kinematic viscosity, while η and η_{eff} are dynamic viscosities.

D Mathematical Foundations

This appendix supplies citation anchors for the analytic notation used in the main chapters and in Appendix B. It separates domain and boundary language, classical function spaces, field notation, differential operators, and norm conventions so later sections can cite only the convention they need.

Convention D.1 (Standing regularity convention). Unless a section states a stronger or weaker hypothesis, classical identities are used only under the regularity needed for their terms to be defined. In particular, fields have the displayed classical derivatives in the indicated domains, boundary traces are defined when boundary terms are written, and integration-by-parts or adjoint identities are invoked only when the required boundary terms are meaningful. When a weak formulation is discussed, the relevant Sobolev space, trace theorem, and boundary regularity assumptions are stated explicitly; Sobolev traces are not inferred from this classical convention. At minimum, boundaries are assumed piecewise C^1 when classical boundary integrals or integration-by-parts identities are used. Outward unit normals are then interpreted on the regular boundary pieces, hence surface-a.e. When a pointwise normal derivative is asserted on all of $\partial\Omega$, the boundary is assumed at least C^1 .

Definition D.2 (Domain and boundary notation). Let $\Omega \subset \mathbb{R}^n$ be a bounded open connected set, with $n \in \mathbb{N}$, and denote its closure by $\overline{\Omega}$. In the hemodynamics settings considered in this report, only $n \in [3]$ is used. The standing regularity convention in Convention D.1 applies whenever traces, outward unit normals, or integration by parts are used. For open Ω ,

$$\overline{\Omega} = \Omega \cup \partial\Omega, \quad \partial\Omega = \overline{\Omega} \setminus \Omega.$$

Since Ω is bounded, $\overline{\Omega}$ is closed and bounded and is therefore compact by the Heine–Borel theorem. The open set Ω itself need not be compact. Since Ω is bounded, $\partial\Omega$ is also compact. When the standing regularity convention supplies a piecewise C^1 boundary, the outward unit normal $\hat{n}$ is interpreted on the regular boundary pieces, hence $dS_{\mathbf{x}}$ -a.e. on $\partial\Omega$.

Definition D.3 (Time intervals and compact space-time sets). For a final time $T_{\text{final}} > 0$, this report uses

$$I_T := (0, T_{\text{final}}), \quad \overline{I}_T := [0, T_{\text{final}}].$$

The open interval I_T is used for interior-time differential identities and space-time PDE regions, while $\overline{I}_T$ is used when initial-time data, endpoint data, sampled time grids, or flow maps are included. When a local final time such as T_* is stated, the same convention applies with T_{final} replaced by that local final time.

For one-dimensional output fields, a compact space-time rectangle is written as

$$K := [0, \ell] \times [0, T_{\text{final}}]$$

or by an explicitly stated local variant. Spaces such as $C(K)$ and $C^1([0, \ell])$ mean continuous functions on the named compact set, with the supremum norm when a norm is written. Spaces such as $L^p(K)$ use the product Lebesgue measure and the same almost-everywhere convention as Definition D.7.

Definition D.4 (Field and multi-index notation). For $\Omega \subset \mathbb{R}^n$, we use the following field notation:

$\phi : \Omega \rightarrow \mathbb{R}$ is a *scalar field*,

$\mathbf{f} : \Omega \rightarrow \mathbb{R}^n$ is a *vector field*,

$\mathbf{T} : \Omega \rightarrow \mathbb{R}^{n \times n}$ is a (*second-order*) *tensor field*,

and we write $\phi(\mathbf{x})$, $\mathbf{f}(\mathbf{x})$, and $\mathbf{T}(\mathbf{x})^1$ for the values at a point $\mathbf{x} \in \Omega$. When referring to a physical quantity, its measurement units are indicated as $[\cdot]$, whereas dimensionless quantities are indicated as $[-]$. The k -th partial derivative of ϕ with respect to coordinate x_i is

$$\partial_{x_i}^k \phi \equiv \frac{\partial^k \phi}{\partial x_i^k}.$$

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$, set $|\alpha| = \alpha_1 + \dots + \alpha_n$. When the derivative exists classically, the corresponding derivative of a scalar field ϕ is

$$D^\alpha \phi \equiv \frac{\partial^{|\alpha|} \phi}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_n^{\alpha_n}}.$$

For $|\alpha| = 1$, $D^\alpha \phi$ is a first-order partial derivative of ϕ , e.g. $\partial_{x_i} \phi$ where $\alpha_i = 1$ and $\alpha_j = 0$ for all $j \in [n] \setminus \{i\}$.

Definition D.5 (Classical function-space notation). Given a domain Ω and a scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, a scalar function space on Ω is an $\mathbb{F}$ -vector subspace of the maps $f : \Omega \rightarrow \mathbb{F}$. Unless a complex-valued space is explicitly specified, this report uses real-valued spaces. Thus $C^0(\Omega)$ denotes the real-valued continuous functions on the open domain Ω . For $k \in \mathbb{N}_0$, $C^k(\Omega)$ denotes functions $\phi : \Omega \rightarrow \mathbb{R}$ such that $D^\alpha \phi$ exists and is continuous in Ω for every $|\alpha| \leq k$. By contrast, $C^k(\overline{\Omega})$ denotes functions in $C^k(\Omega)$ whose derivatives through order k extend continuously to the closure $\overline{\Omega}$. The space $C^\infty(\Omega)$ is the intersection of all $C^k(\Omega)$ for $k \in \mathbb{N}_0$, i.e. the infinitely differentiable functions on Ω . The notation $C^k(\Omega; \mathbb{R}^m)$ denotes the maps $\mathbf{f} : \Omega \rightarrow \mathbb{R}^m$ with components in $C^k(\Omega)$, i.e., $\mathbf{f} = (f_i)_{i \in [m]}$ with $f_i \in C^k(\Omega)$ for all $i \in [m]$.

Definition D.6 (Differential-operator conventions). For real-valued $\phi \in C^1(\Omega)$, the partial derivative with respect to coordinate x_i is

$$\partial_{x_i} \phi = \frac{\partial \phi}{\partial x_i}.$$

¹A second-order tensor value $\mathbf{T}(\mathbf{x}) \in \mathbb{R}^{n \times n}$ can be identified with the matrix of a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ acting on vectors by matrix-vector multiplication.

For $\mathbf{x} \in \Omega$ and a direction $\mathbf{v} \in \mathbb{R}^n$, the directional derivative is

$$D_{\mathbf{v}}\phi(\mathbf{x}) := \langle \nabla\phi(\mathbf{x}), \mathbf{v} \rangle_{\mathbb{R}^n}.$$

If $\phi \in C^1(\overline{\Omega})$, $\mathbf{x} \in \partial\Omega$, and $\hat{\mathbf{n}}$ is the outward unit normal at $\mathbf{x}$, then the boundary normal derivative is

$$\partial_{\hat{\mathbf{n}}}\phi(\mathbf{x}) := \langle \nabla\phi(\mathbf{x}), \hat{\mathbf{n}} \rangle_{\mathbb{R}^n}.$$

For Lebesgue measure $d\mathbf{x}$ on $\mathbb{R}^n$ the integral of ϕ in Ω is

$$\int_{\Omega} \phi(\mathbf{x}) d\mathbf{x}.$$

If $\partial\Omega$ is of class C^1 and $\phi \in C^0(\overline{\Omega})$, then the classical boundary trace $\phi|_{\partial\Omega}$ is continuous on $\partial\Omega$. Since Ω is bounded and $\partial\Omega$ is a compact C^1 hypersurface, this trace belongs to $L^1(\partial\Omega; dS_{\mathbf{x}})$. The corresponding boundary integral is written

$$\int_{\partial\Omega} \phi(\mathbf{x}) dS_{\mathbf{x}}.$$

We use the column-gradient convention for scalar fields. For $\phi \in C^1(\Omega)$, the gradient is

$$\nabla\phi := (\partial_{x_i}\phi)_{i \in [n]} \in \mathbb{R}^n.$$

For $\phi \in C^2(\Omega)$, the Laplacian is

$$\Delta\phi := \nabla \cdot \nabla\phi = \sum_{i \in [n]} \partial_{x_i}^2 \phi.$$

For vector-valued $\mathbf{f} \in C^1(\Omega; \mathbb{R}^n)$ with components $\mathbf{f} = (f_i)_{i \in [n]}$, the vector gradient is the Jacobian with entries $(\nabla\mathbf{f})_{ij} := \partial_{x_j} f_i$. The divergence is the trace of this Jacobian:

$$\operatorname{div} \mathbf{f} := \nabla \cdot \mathbf{f} = \sum_{i \in [n]} \partial_{x_i} f_i = \operatorname{tr}(\nabla\mathbf{f}).$$

For $\mathbf{f} \in C^2(\Omega; \mathbb{R}^n)$, the Laplacian is defined componentwise. The domain integral is also componentwise whenever the components are integrable:

$$\begin{aligned} \Delta \mathbf{f} &:= (\Delta f_i)_{i \in [n]} \\ \int_{\Omega} \mathbf{f}(\mathbf{x}) d\mathbf{x} &:= \left(\int_{\Omega} f_i(\mathbf{x}) d\mathbf{x} \right)_{i \in [n]}. \end{aligned}$$

For tensor-valued $\mathbf{T} \in C^1(\Omega; \mathbb{R}^{n \times n})$ with entries $\mathbf{T} = (T_{ij})_{i,j \in [n]}$, we use the row-divergence convention $(\operatorname{div} \mathbf{T})_i = \sum_{j \in [n]} \partial_{x_j} T_{ij}$. Equivalently, if $\mathbf{t}_i = (T_{ij})_{j \in [n]}$ denotes the i -th row of $\mathbf{T}$, then

$$\begin{aligned} \operatorname{div} \mathbf{T} &:= (\operatorname{div} \mathbf{t}_i)_{i \in [n]} \\ &= (\nabla \cdot \mathbf{t}_i)_{i \in [n]} \\ &= \left(\sum_{j \in [n]} \partial_{x_j} T_{ij} \right)_{i \in [n]}. \end{aligned}$$

For $\mathbf{T} \in C^2(\Omega; \mathbb{R}^{n \times n})$, the Laplacian is defined componentwise. The domain integral is also defined componentwise whenever the entries are integrable:

$$\begin{aligned} \Delta \mathbf{T} &:= (\Delta T_{ij})_{i,j \in [n]} \\ \int_{\Omega} \mathbf{T}(\mathbf{x}) \, d\mathbf{x} &:= \left(\int_{\Omega} T_{ij}(\mathbf{x}) \, d\mathbf{x} \right)_{i,j \in [n]}. \end{aligned}$$

Definition D.7 (C^k and L^p norm conventions). Let $\Omega \subset \mathbb{R}^n$ be bounded and open. We write $C^k(\overline{\Omega})$ for the functions $\phi \in C^k(\Omega)$ such that each derivative $D^\alpha \phi$, $|\alpha| \leq k$, extends continuously to $\overline{\Omega}$. The norm is

$$\|\phi\|_{C^k(\overline{\Omega})} := \sum_{|\alpha| \leq k} \sup_{\mathbf{x} \in \overline{\Omega}} |D^\alpha \phi(\mathbf{x})|.$$

With this norm, $C^k(\overline{\Omega})$ is a Banach space: every Cauchy sequence in $C^k(\overline{\Omega})$ converges, with respect to $\|\cdot\|_{C^k(\overline{\Omega})}$, to a limit in $C^k(\overline{\Omega})$. In particular, for $k = 0$, we have

$$\|\phi\|_{C^0(\overline{\Omega})} := \sup_{\mathbf{x} \in \overline{\Omega}} |\phi(\mathbf{x})| = \max_{\mathbf{x} \in \overline{\Omega}} |\phi(\mathbf{x})|.$$

For continuous functions on compact sets, the max norm agrees with the usual supremum norm.

For $1 \leq p < \infty$, $L^p(\Omega)$ denotes the space of equivalence classes of real-valued measurable functions on Ω , identified up to equality almost everywhere, with finite norm

$$\|\phi\|_{L^p(\Omega)} := \left(\int_{\Omega} |\phi(\mathbf{x})|^p \, d\mathbf{x} \right)^{1/p}.$$

For $p = \infty$,

$$\|\phi\|_{L^\infty(\Omega)} := \operatorname{ess\,sup}_{\mathbf{x} \in \Omega} |\phi(\mathbf{x})|.$$

With this almost-everywhere identification, $L^p(\Omega)$ is a Banach space for $1 \leq p \leq \infty$.

The notation $L^p(\Omega; \mathbb{R}^m)$ denotes equivalence classes of measurable maps $\mathbf{f} : \Omega \rightarrow \mathbb{R}^m$, identified up to

equality a.e., whose components belong to $L^p(\Omega)$, i.e., $\mathbf{f} = (f_i)_{i \in [m]}$ with $f_i \in L^p(\Omega)$ for all $i \in [m]$. When a norm on $L^p(\Omega; \mathbb{R}^m)$ is needed, we use the Euclidean norm in the range. For $1 \leq p < \infty$,

$$\|\mathbf{f}\|_{L^p(\Omega; \mathbb{R}^m)} := \left(\int_{\Omega} \|\mathbf{f}(\mathbf{x})\|_{\mathbb{R}^m}^p d\mathbf{x} \right)^{1/p}.$$

For $p = \infty$, the corresponding convention is

$$\|\mathbf{f}\|_{L^\infty(\Omega; \mathbb{R}^m)} := \operatorname{ess\,sup}_{\mathbf{x} \in \Omega} \|\mathbf{f}(\mathbf{x})\|_{\mathbb{R}^m}.$$

For matrix or second-order tensor values, the pointwise range norm is the Frobenius norm

$$\|\mathbf{T}\|_{\mathrm{F}} := (\mathbf{T} : \mathbf{T})^{1/2} = \left(\sum_{i,j \in [n]} T_{ij}^2 \right)^{1/2}.$$

Thus tensor-valued L^p norms are formed by replacing $|\phi(\mathbf{x})|$ in the scalar L^p formula with $\|\mathbf{T}(\mathbf{x})\|_{\mathrm{F}}$.

E Continuum Derivation Details

This appendix supplements the continuum mechanics details that support the main-body assumptions without interrupting the reduced-order forward-model narrative within this report. It is appendix background for notation and model hierarchy, not additional evidence for the first controlled 1D study. All identities below are classical identities under the regularity needed for the displayed derivatives, traces, and changes of variables. The standing regularity convention, boundary notation, and differential-operator convention are those of Convention D.1, Definition D.2, and Definition D.6. In particular, $\mathbf{div}$ is the row-divergence of a tensor field, and boundary forms assume the stated trace and normal hypotheses; no weak formulation is being asserted in this appendix.

E.1 Material Derivative and Trajectory Details

For a scalar field $\phi \in C^1(\mathcal{Q}_T)$ and a sufficiently regular flow map $\mathcal{L}_t$, define the Lagrangian pullback $\widehat{\phi}(t, \boldsymbol{\xi}) := \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$. Differentiating with respect to time along a trajectory and using $\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \mathbf{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$ gives

$$\frac{d}{dt}\phi(t, \mathcal{L}_t(\boldsymbol{\xi})) = \partial_t \phi(t, \mathbf{x}) + \nabla \phi(t, \mathbf{x}) \cdot \mathbf{u}(t, \mathbf{x}), \quad \mathbf{x} = \mathcal{L}_t(\boldsymbol{\xi}).$$

The identity

$$\operatorname{div}(\phi \mathbf{u}) = \nabla \phi \cdot \mathbf{u} + \phi \operatorname{div} \mathbf{u}$$

also gives

$$D_t \phi = \partial_t \phi + \operatorname{div}(\phi \mathbf{u}) - \phi \operatorname{div} \mathbf{u}.$$

E.2 Jacobian Evolution and Reynolds Transport

Let $\widehat{\mathbf{F}}_t = \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\widehat{J}_t = \det \widehat{\mathbf{F}}_t$. Here $\mathcal{L}_t$ is assumed to be an orientation-preserving C^1 change of variables on the material volume being considered, with the additional time regularity needed to differentiate $\widehat{\mathbf{F}}_t$ and $\widehat{J}_t$. The standard Jacobi identity gives

$$\partial_t \widehat{J}_t = \widehat{J}_t \operatorname{tr} \left(\partial_t \widehat{\mathbf{F}}_t \widehat{\mathbf{F}}_t^{-1} \right) = \widehat{J}_t \operatorname{div} \mathbf{u}(t, \mathcal{L}_t(\boldsymbol{\xi})),$$

which is the Lagrangian form of $D_t J_t = J_t \operatorname{div} \mathbf{u}$ [2, Part I, Ch. 1, Sec. 1.3].

For $V(t) = \mathcal{L}_t(V(0))$, change variables in $I(t) = \int_{V(t)} \phi(t, \mathbf{x}) d\mathbf{x}$:

$$I(t) = \int_{V(0)} \phi(t, \mathcal{L}_t(\boldsymbol{\xi})) \widehat{J}_t(\boldsymbol{\xi}) d\boldsymbol{\xi}.$$

Differentiating under the integral sign and substituting the Jacobian-evolution identity gives

$$I'(t) = \int_{V(t)} (D_t \phi + \phi \operatorname{div} \mathbf{u}) d\mathbf{x} = \int_{V(t)} (\partial_t \phi + \operatorname{div}(\phi \mathbf{u})) d\mathbf{x}.$$

If $V(t)$ has Lipschitz boundary and the trace of $\phi \mathbf{u}$ is integrable on $\partial V(t)$, the divergence theorem gives the boundary form

$$I'(t) = \int_{V(t)} \partial_t \phi d\mathbf{x} + \int_{\partial V(t)} \phi \mathbf{u} \cdot \hat{\mathbf{n}} dS.$$

E.3 Linear Momentum Balance Details

Starting from the material-volume balance

$$\frac{d}{dt} \int_{V(t)} \rho \mathbf{u} dV = \int_{\partial V(t)} \mathbf{T} \hat{\mathbf{n}} dS + \int_{V(t)} \rho \mathbf{f}_b dV,$$

Reynolds transport gives

$$\frac{d}{dt} \int_{V(t)} \rho \mathbf{u} dV = \int_{V(t)} (D_t(\rho \mathbf{u}) + \rho \mathbf{u} \operatorname{div} \mathbf{u}) dV.$$

Using mass conservation, $D_t \rho + \rho \operatorname{div} \mathbf{u} = 0$, this reduces to

$$\int_{V(t)} \rho D_t \mathbf{u} dV = \int_{V(t)} \rho (\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) dV.$$

The surface traction term satisfies

$$\int_{\partial V(t)} \mathbf{T} \hat{\mathbf{n}} dS = \int_{V(t)} \operatorname{div}(\mathbf{T}) dV.$$

This is the tensor row-divergence convention from Definition D.6, applied under the regularity needed for the stress trace $\mathbf{T} \hat{\mathbf{n}}$ and the volume integral of $\operatorname{div} \mathbf{T}$. Since $V(t)$ is arbitrary within the localization class, the differential form is

$$\rho (\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = \operatorname{div}(\mathbf{T}) + \rho \mathbf{f}_b.$$

The conservative Eulerian form

$$\partial_t(\rho \mathbf{u}) + \mathbf{div}(\rho \mathbf{u} \otimes \mathbf{u}) = \mathbf{div}(\mathbf{T}) + \rho \mathbf{f}_b$$

is equivalent whenever the continuity equation holds, because

$$\partial_t(\rho \mathbf{u}) + \mathbf{div}(\rho \mathbf{u} \otimes \mathbf{u}) = \rho (\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) + \mathbf{u} (\partial_t \rho + \mathbf{div}(\rho \mathbf{u})) .$$

F Navier–Stokes Coordinate and Energy Details

This appendix collects mathematical orientation material for the 3D Navier–Stokes model. It is background for the continuum hierarchy; it is not evidence for the 1D study specification or any downstream validation claim. All formulas are read under the standing regularity and differential-operator conventions in Appendix D. In particular, $\mathbf{div}$ is row-divergence for tensor fields, $\Delta \mathbf{u}$ is the componentwise vector Laplacian, tensor norms are Frobenius norms when written, and η is dynamic viscosity while $\nu = \eta/\rho$ is kinematic viscosity as in Remark C.3.

F.1 Viscous-Term Simplification and Energy Context

For a Newtonian fluid with constant dynamic viscosity η and sufficiently smooth velocity field,

$$\mathbf{div}(2\eta \mathbf{D}(\mathbf{u})) = \eta \mathbf{div}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top) = \eta \Delta \mathbf{u} + \eta \nabla(\mathbf{div} \mathbf{u}).$$

For incompressible flow, $\mathbf{div} \mathbf{u} = 0$, so

$$\mathbf{div}(2\eta \mathbf{D}(\mathbf{u})) = \eta \Delta \mathbf{u}, \quad \rho^{-1} \mathbf{div}(2\eta \mathbf{D}(\mathbf{u})) = \nu \Delta \mathbf{u}, \quad \nu = \eta/\rho.$$

This term is the momentum-diffusion contribution; it does not disappear in the incompressible Newtonian limit.

Classical local well-posedness results for Navier–Stokes depend on the domain, boundary conditions, forcing, initial-data space, and solution class. Under sufficiently smooth compatible data one obtains a unique strong solution on a maximal time interval. If the maximal interval is finite, continuation fails through blow-up of the norm controlled by the chosen well-posedness theory.

F.2 Cylindrical Coordinate Form

For (r, θ, z) with orthonormal basis $(\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z)$,

$$x = r \cos \theta, \quad y = r \sin \theta, \quad \mathbf{u} = u_r \mathbf{e}_r + u_\theta \mathbf{e}_\theta + u_z \mathbf{e}_z.$$

The component formulas are local cylindrical-coordinate formulas on patches where $r > 0$. Axis regularity, wall traces, and boundary conditions require separate hypotheses and are not supplied by the coordinate

notation alone. The differential operators used in cylindrical divided-by-density Navier–Stokes forms are

$$\begin{aligned}(\mathbf{u} \cdot \nabla) &= u_r \partial_r + \frac{u_\theta}{r} \partial_\theta + u_z \partial_z, \\ \nabla p &= \partial_r p \mathbf{e}_r + \frac{1}{r} \partial_\theta p \mathbf{e}_\theta + \partial_z p \mathbf{e}_z, \\ \operatorname{div} \mathbf{u} &= \frac{1}{r} \partial_r(r u_r) + \frac{1}{r} \partial_\theta u_\theta + \partial_z u_z.\end{aligned}$$

The rate-of-deformation components in the cylindrical orthonormal basis are

$$\begin{aligned}D_{rr} &= \partial_r u_r, & D_{\theta\theta} &= \frac{1}{r} \partial_\theta u_\theta + \frac{u_r}{r}, & D_{zz} &= \partial_z u_z, \\ D_{r\theta} &= \frac{1}{2} \left(\partial_r u_\theta - \frac{u_\theta}{r} + \frac{1}{r} \partial_\theta u_r \right), & D_{rz} &= \frac{1}{2} (\partial_r u_z + \partial_z u_r), & D_{\theta z} &= \frac{1}{2} \left(\frac{1}{r} \partial_\theta u_z + \partial_z u_\theta \right).\end{aligned}$$

Use the scalar shear-rate convention $\dot{\gamma} = \sqrt{2\mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{u})}$. Then

$$\tau_{ij} = 2\eta_{\text{eff}}(\dot{\gamma}) D_{ij}, \quad i, j \in \{r, \theta, z\}.$$

When a source writes the law as a function of $D : D$ or $|D|_{\mathbb{F}}^2$, this report translates it through this convention, consistent with Remark C.3. Writing $\mathbf{f} = f_r \mathbf{e}_r + f_\theta \mathbf{e}_\theta + f_z \mathbf{e}_z$, the momentum components are

$$\begin{aligned}(\text{radial}) \quad & \partial_t u_r + u_r \partial_r u_r + \frac{u_\theta}{r} \partial_\theta u_r + u_z \partial_z u_r - \frac{u_\theta^2}{r} \\ &= -\frac{1}{\rho} \partial_r p + \frac{1}{\rho} \left[\frac{1}{r} \partial_r(r \tau_{rr}) + \frac{1}{r} \partial_\theta \tau_{r\theta} + \partial_z \tau_{rz} - \frac{1}{r} \tau_{\theta\theta} \right] + f_r, \\ (\text{azimuthal}) \quad & \partial_t u_\theta + u_r \partial_r u_\theta + \frac{u_\theta}{r} \partial_\theta u_\theta + u_z \partial_z u_\theta + \frac{u_r u_\theta}{r} \\ &= -\frac{1}{\rho r} \partial_\theta p + \frac{1}{\rho} \left[\frac{1}{r} \partial_r(r \tau_{r\theta}) + \frac{1}{r} \partial_\theta \tau_{\theta\theta} + \partial_z \tau_{\theta z} + \frac{1}{r} \tau_{r\theta} \right] + f_\theta, \\ (\text{axial}) \quad & \partial_t u_z + u_r \partial_r u_z + \frac{u_\theta}{r} \partial_\theta u_z + u_z \partial_z u_z \\ &= -\frac{1}{\rho} \partial_z p + \frac{1}{\rho} \left[\frac{1}{r} \partial_r(r \tau_{rz}) + \frac{1}{r} \partial_\theta \tau_{\theta z} + \partial_z \tau_{zz} \right] + f_z.\end{aligned}$$

The incompressibility condition is

$$\frac{1}{r} \partial_r(r u_r) + \frac{1}{r} \partial_\theta u_\theta + \partial_z u_z = 0.$$

G Numerical Methods Details

This appendix supplies the numerical-method detail behind Sec. 1.2.3 and the comparison boundary in Sec. 1.2.4. It is supporting method detail, not a claim that the current solver has external-solver parity. The continuous identities are read under Convention D.1; computed quantities are read through the following numerical-record convention.

Convention G.1 (Computed realization record). A computed realization is interpreted only through the model, closure, discretization, grid, timestep policy, boundary-state rule, output operator, and diagnostic metadata recorded for that realization. Symbols such as D_t , D_z , numerical fluxes, residual arrays, and sampled norms denote recorded discrete objects unless a section explicitly switches back to a continuum identity.

The time interval, output rectangle, and norm notation follow Definition D.3 and Definition D.7; discrete residuals use the recorded D_t , D_z , grid, and quadrature weights rather than unnamed continuum derivatives.

G.1 Elastic-potential balance-law form

The reduced state is $\mathbf{U} = (A, Q)^\top$. For the classical elastic balance-law contract, the reference elastic-potential form is

$$\begin{aligned}\partial_t A + \partial_z Q &= 0, \\ \partial_t Q + \partial_z \left(\frac{Q^2}{A} + \Psi(A, z, t) \right) &= S_{\text{geom}}(A, z, t) - C_f(z, A, Q) \frac{Q}{A}.\end{aligned}$$

The elastic potential satisfies

$$\partial_A \Psi(A, z, t) = \frac{A}{\rho} \partial_A p_g(A, z, t),$$

and the geometry source is

$$S_{\text{geom}}(A, z, t) = \partial_z \Psi(A, z, t)|_A - \frac{A}{\rho} \partial_z p_g(A, z, t)|_A.$$

For the baseline wall law, assuming p_{ext} is independent of A ,

$$c^2(A, z) = \frac{A}{\rho} \partial_A p_g(A, z, t) = \frac{\beta(z) \sqrt{A}}{2\rho A_0(z)}.$$

The strictly hyperbolic positive-area region for this classical model has wave speeds

$$\lambda_{\pm}(A, Q; z) = \frac{Q}{A} \pm c(A, z).$$

The boundary-state construction is intended for the subcritical case $\lambda_- < 0 < \lambda_+$, so one datum enters at each end of the vessel. The displayed derivatives of A_0 , β , p_g , and Ψ are continuum notation for the model contract. A computed record may realize them through explicitly recorded finite-difference or source-split operators; changing that realization changes the computed map.

G.2 Current semi-discrete solver operator

The current Canic extended implementation uses the code variable A for a radius-squared area coordinate. In this subsection write that coordinate as $a = R^2$. The physical circular cross-sectional area is $A_{\text{phys}} = \pi a$. This is a local convention for the active solver record; the continuum and classical 1D model sections keep A for physical cross-sectional area.

For N finite-volume cells, let

$$z_i = \left(i - \frac{1}{2}\right) \Delta z, \quad \Delta z = \frac{L}{N}, \quad u_h(t) = (a_1(t), \dots, a_N(t), Q_1(t), \dots, Q_N(t))^\top.$$

The semi-discrete system exposed by `semidiscretize` and `rhs!` is

$$\dot{u}_h = \mathcal{L}_h(u_h, t; \theta),$$

where θ denotes the recorded physical, geometry, grid, and solver parameters. Componentwise,

$$\begin{aligned} \dot{a}_i &= -\frac{\hat{F}_{i+1/2}^a - \hat{F}_{i-1/2}^a}{\Delta z}, \\ \dot{Q}_i &= -\frac{\hat{F}_{i+1/2}^Q - \hat{F}_{i-1/2}^Q}{\Delta z} + \hat{S}_i. \end{aligned}$$

All divisions by a in the implementation use the recorded positive-area floor.

With $K = Eh/(1 - \sigma^2)$ and $\alpha_c(R'_0) = -2(R'_0)^2/35$, the current full-state flux is

$$\mathbf{F}(a, Q, z) = \left(Q, (\alpha + \alpha_c(R'_0(z))) \frac{Q^2}{a} + \frac{K}{3\rho R_{\max}^2} a^{3/2} \right)^\top.$$

At each interface,

$$\hat{\mathbf{F}}_{i+1/2} = \frac{1}{2} \left[\mathbf{F}(\mathbf{U}_{i+1/2}^-, z_{i+1/2}) + \mathbf{F}(\mathbf{U}_{i+1/2}^+, z_{i+1/2}) \right] - \frac{1}{2} s_{i+1/2} \left(\mathbf{U}_{i+1/2}^+ - \mathbf{U}_{i+1/2}^- \right),$$

where $\mathbf{U} = (a, Q)^\top$ and the present implementation uses neighboring first-order states plus ghost boundary

states. The recorded local speed is

$$s_{i+1/2} = \max \left\{ s_h(\mathbf{U}_{i+1/2}^-, z_{i+1/2}), s_h(\mathbf{U}_{i+1/2}^+, z_{i+1/2}) \right\},$$

with

$$\begin{aligned} s_h(a, Q, z) &= \max \{ |\alpha_{\text{eff}} u - \chi|, |\alpha_{\text{eff}} u + \chi| \}, \\ \alpha_{\text{eff}} &= \alpha + \alpha_c(R'_0(z)), \quad u = \frac{Q}{a}, \\ \chi^2 &= (\alpha_{\text{eff}} u)^2 - \alpha_{\text{eff}} u^2 + \frac{K}{2\rho R_{\text{max}}^2} \sqrt{a}. \end{aligned}$$

The source term uses the recorded centered interior and one-sided boundary difference D_z^h on the cell averages. Let $g_\alpha = \alpha/(\alpha - 1)$ and $\alpha'_c(z) = -4R'_0(z)R''_0(z)/35$. The implemented source is

$$\begin{aligned} \widehat{S}_i &= -2\nu g_\alpha \frac{Q_i}{a_i} + \frac{K}{\rho R_{\text{max}}^2} a_i R'_0(z_i) - a_i \widehat{\partial_z p_{2i}} + \frac{Q_i^2}{a_i} \alpha'_c(z_i), \\ \widehat{\partial_z p_{2i}} &= \nu g_\alpha \left[\frac{Q_i R''_0(z_i)}{a_i R_0(z_i)} - \frac{Q_i (R'_0(z_i))^2}{a_i R_0(z_i)^2} + \frac{(D_z^h Q)_i R'_0(z_i)}{a_i R_0(z_i)} - \frac{Q_i (D_z^h a)_i R'_0(z_i)}{a_i^2 R_0(z_i)} \right]. \end{aligned}$$

G.3 Time and backend realization

The native backend advances the semi-discrete operator with the recorded third-order strong-stability-preserving Runge–Kutta map

$$u_h^{n+1} = \Phi_{\Delta t_n}^{\text{SSPRK3}}(u_h^n; \mathcal{L}_h).$$

In expanded form, using Π for the recorded positive-area projection on the a -components,

$$\begin{aligned} u^{(1)} &= \Pi(u^n + \Delta t_n \mathcal{L}_h(u^n, t^n)), \\ u^{(2)} &= \Pi\left(\frac{3}{4}u^n + \frac{1}{4}\left[u^{(1)} + \Delta t_n \mathcal{L}_h(u^{(1)}, t^n + \Delta t_n)\right]\right), \\ u^{n+1} &= \Pi\left(\frac{1}{3}u^n + \frac{2}{3}\left[u^{(2)} + \Delta t_n \mathcal{L}_h(u^{(2)}, t^n + \Delta t_n)\right]\right). \end{aligned}$$

The native step size is

$$\Delta t_n = \min \left\{ \Delta t_{\text{max}}, \frac{\text{CFL} \Delta z}{\max_i s_h(a_i^n, Q_i^n, z_i)}, T_{\text{final}} - t^n \right\}.$$

The SciML backend constructs an `ODEProblem` for the same $\dot{u}_h = \mathcal{L}_h(u_h, t; \theta)$ and applies the selected `OrdinaryDiffEq` policy recorded in the solve specification. Thus backend comparisons change the time integrator, not the semi-discrete spatial operator.

G.4 Boundary-state realization

The current finite-volume record imposes one incoming datum at each boundary in the subcritical regime using the code's Riemann-invariant construction. Let

$$c_0 = \left(\frac{K}{2\rho R_{\max}^2} \right)^{1/2}.$$

At the inlet, the incoming flow is $Q_{\text{in}} = R_0(0)^2 \bar{u}_{\text{in}}$. The outgoing invariant from the first interior cell is

$$w_- = \frac{Q_1}{a_1} - 4c_0 a_1^{1/4},$$

and the ghost coordinate a_{in} solves

$$\frac{Q_{\text{in}}}{a_{\text{in}}} - 4c_0 a_{\text{in}}^{1/4} = w_-.$$

At the outlet, the radius-squared coordinate is fixed to $a_{\text{out}} = R_0(L)^2$. The outgoing invariant from the last interior cell is

$$w_+ = \frac{Q_N}{a_N} + 4c_0 a_N^{1/4},$$

and the ghost flow is

$$Q_{\text{out}} = a_{\text{out}} w_+ - 4c_0 a_{\text{out}}^{5/4}.$$

The boundary update therefore constrains the incoming information and leaves the outgoing information tied to the interior state. Boundary-quality plots and reflected-wave sidecars are diagnostics until explicit thresholds are admitted.

G.5 Residual diagnostics

Stored residuals are post-processing diagnostics on sampled fields, not proof of the continuous PDE residual. The symbols D_t and D_z in this subsection denote the recorded temporal and spatial finite-difference operators on the saved grid, not weak derivatives. For a stored field sequence, the semi-discrete defect is read as

$$\mathcal{E}_h^n = D_t u_h^n - \mathcal{L}_h(u_h^n, t^n; \theta).$$

Componentwise, this gives

$$\mathcal{E}_i^a(t^n) = D_t a_i^n - \mathcal{L}_{h,i}^a(u_h^n, t^n; \theta), \quad \mathcal{E}_i^Q(t^n) = D_t Q_i^n - \mathcal{L}_{h,i}^Q(u_h^n, t^n; \theta).$$

The E1 diagnostic summary reports these residuals on reviewed interior cells with recorded finite-field, positivity, CFL, and pressure-drop sign checks. The numbers support the dataset contract and local consistency reading; they do not establish a convergence theorem or external parity. Any residual norm or error norm reported from these arrays is a sampled norm tied to the numerical record, rather than an unqualified continuum $L^p(K)$ norm.

H Code Availability and Build Statement

This report is built from the local L^AT_EX sources rooted at `final-report.tex`. The compiled PDF should be treated as the rendered form of those sources. Any numerical-study code, solver configuration, or generated data used later in the report should be cited in the relevant study section with enough information to identify the method, inputs, and output records.

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